



Indira Gandhi National Open University
School of Journalism and New Media Studies

MDC-005

**DEVELOPMENT: INFORMATION AND
COMMUNICATION TECHNOLOGIES**

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COMMUNICATION TECHNOLOGIES**



**School of Journalism and New Media Studies
Indira Gandhi National Open University
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MDC-005 INTRODUCTION TO COURSE: DEVELOPMENT INFORMATION AND COMMUNICATION TECHNOLOGIES

Block No. 1: ‘Understanding ICT’ consists of 4 units, these are Internet as a Medium, Digital Media & Society, Issues of Access and Participation and Policy Frameworks and Regulations.

Unit 1: Internet as a Medium describes the complex structure of cyberspace, that is a platform through which internet-based communication takes place between users for various purposes. There are multiple opportunities for users to communicate via computers or mobile gadgets between one to one, or one to many or many to many or many to one. The speciality with this medium is that one medium can be used for personal purposes as well as for public communication exchanges as depending on the situation and context the role changes. Here in cyberspace, all the users have equal access to information at the same time, they have the capability to participate in the information exchanges. Interestingly all the communication through cyberspace is by default reaches every nook and corner of the globe, transcending multiple boundaries. Users can assemble in one place based on their interest, cutting across their geographical locations and time, and form online communities to interact and exchange information on a given topic/interested areas. In this Unit, you will explore all such scenarios with the inputs coming from various scholarly resources.

Unit 2: Digital Media & Society In this Unit, you will learn about the relationship between the digital media and social, cultural, political and economical realms of the society.

The information exchanges through cyberspace also produce unique relationships and consequences that you will learn in this Unit with the help of multiple theoretical perspectives.

Unit 3: Issues of Access and Participation explains the processes involved in the Information and Communication Technology and its infrastructure diffusion through the Indian social context. Every communication medium needs infrastructure and platform to operationalise the information exchanges. Here in the digital medium it is based on technology, more complex advanced set of gadgets, technological skills and competencies are required. Hence the varying social and economic background creates a gap while adapting to this technology-based communication platform. This gap is being referred to as digital inequality. The same has been initially termed as digital divide - the binary between those who are having access to the digital medium and those who are not having access to it. In this Unit, you will learn about the diffusion of ICTs both in Indian context as well as from the global perspective. Also conceptual details on digital inequality, efforts taken to

bridge the gap between the 'haves' and 'have nots' from the Indian experiences are given in a detailed manner.

Unit 4: Policy Frameworks and Regulations gives a detailed description of the massive expansion of the Internet and mobile users in the Indian context, the various services provided through this medium, and the agencies involved in establishing between IT infrastructure. And more importantly it describes how this medium is being regulated through various policies, regulations and laws. Right from the National Telecom Policy of 1994 till National Digital Communications Policy of 2018, this Unit traces the policy guidelines as well as targets which need to be achieved in the IT as well as ICT sectors by 2022.

Block No. 2: 'ICT: Conceptual Framework' comprises four units, ICT for Development – An Overview, E- Governance; Policy and Framework, E- Governance in Rural Development and E- Governance in Urban Development.

Unit 5: ICTS for Development – An Overview focuses on meaning and attributes of ICT, interface between ICT and development of various sectors and edevelopment strategies.

Unit 6: E- Governance: Policy and Framework enumerates the concept, various stages, models of e-governance. ICT based digital media provides immense opportunities to the government departments and public institutions to provide public service deliveries through the online mode and to reduce the barriers faced by citizens to get the full benefits of various welfare schemes and government services. In this Unit, it is explained that if the full potential of ICT is reaped appropriate for governance purposes, it will give the much-needed support to the economic development. Besides that, challenges and opportunities from the Indian perspective are also explained in this Unit.

Unit 7: E- Governance in Rural Development deals with the meaning and importance of e-governance, the interrelation between e-governance and rural development, various dimensions of the digital divide, models, perspectives and cases of e-governance in rural development. Bringing the modernisation and improvements in the infrastructure in remote and rural areas is a challenging phenomenon, mainly due to the digital inequality in those areas. In this Unit, the digital divide's political, social and cultural dimensions are critically analysed from the Indian perspectives.

Unit 8: E- Governance in Urban Development deals with the importance of e-governance in urban development, international and national initiatives of e-governance in urban development and challenges in e-governance. The examples and case studies of using Information and Communication Technology in development sectors, particularly in facilitating social schemes, are discussed elaborately in this Unit.

Block No. 3: 'ICT: Approaches and Application' consists of three units, ICT for Education, ICT for Health and ICT for Disability.

Unit 9: ICT for Education describes the nuances of integrating ICT tools for teaching and learning systems. This Unit traces the Indian experiences of e-learning along with issues and challenges of e-learning opportunities for Indian educational institutions. Standalone online courses are gaining momentum at recent times and the significance of Online and Open Distance Learning/Life-long learning is increasing. By integrating ICT in its delivery methods it is hoped to resolve the last-mile connectivity issues.

Unit 10: ICT for Health explain the access, understanding and participation in health communication. This Unit explains the concept of media and information literacy for utilising ICT tools by an individual or a community's public health care system. Public health care system and its quality determines the human living conditions. The ICT tools allow an individual or a community to interact with information providers to seek and utilize the information for health benefits. There are multiple factors which determine the media health literacy effectiveness, particularly the age, educational level, technological understanding, economical affordability and social and cultural settings which play a critical role in it. This Unit explains the concept of media health literacy with the help of a few theoretical frameworks.

Unit 11: ICT for Disability deals with the technological intervention to empower People with Disabilities to overcome their barriers to reap various services benefits. In this Unit, two major components are discussed - suitability of ICT for PwD and intervention efforts for PwD. Under the suitability aspects, the Unit is describing the benefits and emerging trends of ICT for disability. Concerning intervention efforts, it discusses the measures taken by the institutional initiatives as well as individual actions.

Block No. 4: 'India's Tryst with Digital Age' consists of four units, Internet and Marginalized Sections, Participatory Online Media, Online Activism and ICT for ODL.

Unit 12: Internet and Marginalized Sections is about how the Internet is dealing with the marginalised sections of the society. In this Unit, the process of marginalisation is discussed along with the representation of some groups on the Internet. The very nature of interactivity should have been exploited extensively in order to give a voice to the voiceless – the strengths and weaknesses of this statement are being discussed in this Unit.

Unit 13: Participatory Online Media describes multiple levels of engagements of users with the Internet. More specifically, this Unit focuses on the civic engagement of users regarding the common issues of nearby communities, particularly with reference to young users, and it is explained in detail from Jenkins's Participatory Culture perspective. The concept of prosumers and produsers are explored in this Unit in detail as the Internet user takes a dual role as a receiver of information and at the same time s/he participates and contributes to various forms of media elements.

Unit 14: Online Activism gives a broad idea of how Internet resources are being utilised by individuals and organisations towards collective efforts. The transcending nature of the Internet's reach brings people together to change a policy or to voice their concerns about the need for a new social order. These collective efforts may be against political issues or common society affairs or for humanity as a whole. For these very purposes of social movements for the betterment of society and the world, the Internet resources are being used to mobilise the support in terms of opinions and finances that helps the groups/movements to translate their online efforts into offline actions. The whole concept of online activism is dealt elaborately in this Unit.

Unit 15: ICT for ODL explores the various available technological tools to integrate with Open and Distance Learning. It traces the historical dimensions of the integration process of Information and Communication Technology (ICT) with the ODL and contextualises it with the various generational shifts in this educational system. More importantly, this Unit enumerates the benefits of ICT for ODL and explores the avenues of integration from the futuristic perspectives.

Block No. 5: 'Knowledge Society and Digitalisation' consists of three units, these are Dimensions of Knowledge Society: Access and Equity Issues, Democracy and Digital Media and ICT and Knowledge Society: Challenges & Opportunities.

Unit 16: Dimensions of Knowledge Society: Access and Equity Issues deals with meaning, features and basis of the emerging knowledge society. One of the basic features of the current pace of globalization is its primary importance of ICTs and the emergence of ICT led knowledge society. Here knowledge becomes a commodity and it becomes essential to possess knowledge to achieve success in life and to accumulate wealth. The first half of the unit deals with the meaning and features of knowledge society and its various aspects. Then it goes on analyzing the three basic of knowledge society, I.E. generation, dissemination and deployment of Knowledge, in terms of skill required for knowledge generation, infrastructure for knowledge dissemination and employment structure that relates to knowledge deployment.

Unit 17: Democracy and Digital Media explains other forms of participation of Internet users in the field of democracy. While explaining the process of participation in democracy, this Unit explains different levels of engagements and the relationship between digital media and democracy. Apart from this, the Unit cites examples from India as well as from the global context.

Unit 18: ICT and Knowledge Society: Challenges & Opportunities looks into the sociological criticisms in empirical and theoretical terms of the concept of knowledge society. The knowledge society is a concept that has been criticized conceptually, theoretically and empirically. Although the unit deals with each of these criticisms, certain theoretical and empirical criticisms are focused on.

Block**1****UNDERSTANDING ICT**

UNIT 1**Internet as a Medium**

UNIT 2**Digital Media & Society**

UNIT 3**Issues of Access and Participation**

UNIT 4**Policy Frameworks and Regulations**

UNIT 1 INTERNET AS A MEDIUM

Structure

- 1.0 Introduction
- 1.1 Learning Outcomes
- 1.2 Internet as a Medium of Communication
 - 1.2.1 Conceptual Framework of Cyberspace
 - 1.2.2 Functional Dimensions of Cyberspace
 - 1.2.3 Characteristics of Cyberspace
- 1.3 Types of Internet-Based Communication
 - 1.3.1 Dynamics of Communication Process in CMC
 - 1.3.2 Cohesive Force of Online Group Communication
- 1.4 Forms of Computer-Mediated Communication
 - 1.4.1 Impersonal CMC
 - 1.4.2 Interpersonal CMC
 - 1.4.3 Hyper personal CMC
- 1.5 Virtual Communities
- 1.6 Let Us Sum Up
- 1.7 Keywords
- 1.8 References and Further Readings
- 1.9 Check Your Progress: Possible Answers

1.0 INTRODUCTION

In the previous courses, you have learnt characteristics of various mass media entities such as the newspaper, radio, television, etc. In the case of newspapers, it has a limited lifecycle, and there is a geographical boundary in terms of its physical reach to its intended readers. Similarly, the electronic media - radio and television - have their limitations of reach of satellite footprints. However, the electronic media is better placed, as compared to the print media. On the other hand, we now have a medium that transcends all these boundaries and limitations along with a worldwide reach and irrespective of geopolitical governance. The Internet as a medium is emerging as a dominant one in the mass media milieu. Interestingly, the Internet has multiple dimensions - you can use it for your personal interactions as well as for professional tasks.

Every mass media needs a channel and a medium. The channel is the carrier while the medium facilitates the communication exchange process. In the case of the newspaper - print is the channel and news stories of newspapers are the medium, in electronic media - electronic waves are channels, and programmes of electronic media are its medium. Similarly, bandwidth and broadband are channels for the Internet, and information pockets are its medium.

New Media and Society In this Unit, you will learn about the concept of cyberspace, and the basic structure of the Internet through which all the digital/online communication process occurs and computer-mediated communication - a term used for the types of communication associated with the Internet-based communication. One of the unique qualities of Internet communication is its ability to gather several users into specific areas - not physically, but gatherings are based on common interest. Such gatherings are called virtual communities. You will learn about the multiple communication dimensions of the Internet in this Unit.

1.1 LEARNING OUTCOMES

After reading this Unit, you will be able to:

- understand the complex structure of cyberspace;
- explain the different forms of computer-mediated communication;
- state diverse patterns of Internet-based communication; and
- discuss the various dimensions of virtual/online communities.

1.2 INTERNET AS A MEDIUM OF COMMUNICATION

Information and Communication Technology (ICT) has given us a powerful communication tool through which we can merge many of our information and dissemination needs. About two decades back, we had only the mainstream media outlets for our society-wide communication purposes, which are still mostly vertical, i.e. we receive the messages in a top-down model and in which we have limited options of participation. The ICT platform has revolutionised the way we communicate with one another as well as with society at large. Theoretically, we can call this platform as cyberspace, and it includes all dimensions of digital and online media and its platforms. Cyberspace provides a horizontal model of communication, and there is no hierarchy between the users. Every user can access the information, and at the same time, each user has the facility to communicate, not just with friends or relatives or colleagues, but to the entire globe. Such titanic changes in our communication platforms need to be understood academically.

1.2.1 Conceptual Framework of Cyberspace

Through computer or mobile, we usually immerse ourselves in various activities, and the platform through which we are working and interacting is called cyberspace. This term is not a new one. It was coined way back in 1984 by the science fiction writer William Gibson in his novel *Neuromancer*. While explaining the new term cyberspace, Gibson stated that “A consensual hallucination...a graphic representation of data abstracted from the back of every computer in the human system. Unthinkable complexity lines of light ranged in the non-space of the mind, clusters, and constellations of data. Like city lights receding.” We will see yet another definition from a famous

scholar and pioneer in online communities studies, Harold Rheingold. Rheingold's (1995) definition of cyberspace is "Conceptual space where words, human relationships, data, wealth, and power are manifested by people using CMC (Computer-mediated Communication) technology."

The tools of ICT has created a platform for billions of people to gather Internet as a Medium and interact with each other. The core characteristics of fruitful interactions between humans are the primary component of cyberspace. One of the earlier scholars of cyberspace, Loader (1997) explains its unique features. He lists its characteristics as follows: "A computer-generated public domain which has no territorial boundaries or physical attributes and is in perpetual use. To date, its most potent manifestation is that matrix of electronic telecommunication and computer networks, usually referred to as the Internet, which links millions of people globally which is growing at a rapid rate daily and is taking new shape and direction as a consequence of the voluntary actions of its participants, and it is claimed that is not controlled by any single authority."

Communication through cyberspace is an extension of offline human-to human interactions. We leave our physical presence, and we take different avatars in the online space. All these interactions are performed in the ICT enabled and simulated environment.

In the virtual space, people still meet face-to-face but under new definitions of "meet" and "face." The cyber platform is the passage where physically separated people are bound together by common beliefs and practices. The essential element in cyberspatial social relations is the sharing of information. It is not sharing in the sense of transmission of information but binding communities in cyberspace. It is the ritual of information sharing that pulls it together.

Rather than incidental meetings, collective and common interest fuels the process of communication in cyberspace. Even before the invention of the Internet, Licklider and Taylor (1968) predicted that: "In most fields, they will consist of geographically separated members, sometimes grouped in small clusters, and sometimes working individually. There will be communities not of common location, but of common interest. In each geographical sector, the total number of users will be large enough to support extensive general-purpose information processing and storage facilities. Life will be happier for the individual who is in online because the people with whom he/she interacts most strangely will be selected more by the commonality of interests and goals than by accidents of proximity."

Interactivity is an important element of new media. It eliminates the time barrier for a user to access the contents conveniently anytime anywhere. The time-shifting facility enhances a user's involvement in the online environment, and time becomes a crucial factor here. The virtual platform offers not only a multitude of topic areas but also the ability to participate at a convenient time. There is no weekly meeting to catch up after work. The meetings happen whenever the users have time to log in and read the new postings. This time-shifting allows users time to ponder upon a particularly

serious posting or article and write a coherent response. People from many time zones can participate in a discussion, and they have more control over their online experience, which allows them to participate at their convenience.

1.2.2 Functional Dimensions of Cyberspace

Cyberspace guarantees users the freedom to move along with the medium, which has dimensionality, continuity, curvature, density, and limits. It is a socially constructed and reconstructed space and essentially a re-conceived public sphere for social, political, economic, and cultural interaction. Cyberspace provides a new arena for public life in which a user can adopt any meaningful role like an author, public rhetorician, statesman, and pundit, which is uncommon in the mass communication process. In cyberspace, says Benedikt (1991), “The common man and the information worker-cowboy or infocrat – can search, manipulate, create or control information directly; he can be entertained or trained, seek solitude or company, win or lose power indeed, can “live” or “die” as he will.” Cyberspace has two dimensions concerning the openness of its communication activity. It is “public space, at the same time it is private also,” contends Fernback (1997), “where via email, two users can argue politics or fall in love, or several users on a private list server can strategise a meeting or discuss the finer points of a classroom lecture.”

The CMC users’ relationships can range from the cold, professional encounter, to the hot, intimate rendezvous. Rheingold (1995), while describing routine activities of this medium says a user can “Exchange pleasantries and argue, engage in intellectual discourse, conduct commerce, exchange knowledge, share emotional support, make plans, brainstorm, gossip, feud, fall in love, find friends and lose them, play games, flirt, create a little high art and even engaged in a lot of idle talks. People ... do just about everything people do in real life, but we leave our bodies behind. You cannot kiss anybody, and nobody can punch you in the nose, but a lot can happen within those boundaries. To the millions who have been drawn into it, the richness and vitality of computer-linked cultures are attractive, even addictive.”

The cyberspace is a multi-faceted environment where one can find immensely varied, amalgamated, and combined social relations. Fernback (1997) gives a detailed account on the inner view of cyberspace: “It is a repository for collective memory - it is popular culture, it is narrative created by its inhabitants that remind us who we are, it is life as lived and reproduced in pixels and virtual texts. It is sacred and profane, and it is workspace and leisure space, it is a battleground and a nirvana, it is real, and it is virtual, it is ontological and phenomenological. Cyberspace is an arena of power; CMC users act every day on the assumption that the tyranny of geography can be overcome within cyberspace. It is smaller, more intimate, and almost more imaginable than “the public,” which can no longer fit into a stadium.”

The revolution in global networked communications has given rise to a new generation of social technologies, including mechanisms for the formation

and cultivation of interpersonal relationships. The global reach of the Internet not only facilitates communication among members of existing distributed groups and teams but also provides a medium for the formation and cultivation of new relationships by providing virtually instantaneous access to thousands of potential contacts that have compatible interests and spheres of expertise.

The urbanisation and industrialisation have resulted in a mass society in the urban context where the social relationship is highly fragmented. CMC is acting like a new cohesive force to build an online relationship. As Jones (1998) argues, “CMC allows us to customise our social contacts from fragmented communities and to plan, organise and make efficient and Internet as a Medium social contacts.” Linda Harasm (1993) found that social communication is a primary component of CMC and is well able to organise thoughts about the use of CMC around social situations, rather than working areas.

Even though CMC facilitates tremendous information exchange power, it has its inadequacies. One prominent negative feature, cited by most CMC scholars, is that CMC occurs in cyberspace mostly with letters and words. This textual communication filters out demographic and socio-economic information about the user, such as sex and social status, which also limits relational meaning.

1.2.3 Characteristics of Cyberspace

In comparison with face-to-face communication, which is considered as the communication standard against which all others are found to be inferior, a CMC user cannot hear intonation that signals a joke or see puzzled expressions that convey confusion. Also, users often face problems in coordinating informal discussions owing to the lack of information feedback, the absence of social influence cues, and depersonalisation due to the lack of non-verbal involvement. Research on spoken conversational interaction shows that simultaneous feedback plays a vital role in signalling listenership, timing turn-taking effectively, and maintaining continuous interaction. Also, the absence of simultaneous feedback may result in discontinuity and/or overlap within turn sequences, as well as generally making it more difficult for message producers to tailor their messages to respond to recipients' interests and needs. When the relevant responses disrupt, CMC users may experience difficulty in tracking sequential exchanges, and as a result, the interaction may become fragmented.

To theorise this impersonal communication pattern in 1987, Culnan and Markus described it as the “cues filtered out” theory (also called “reduced cues”), which posits that the computer has a “low social presence” because it filters out important aspects of communication that participants in face to-face communication are privy to (paralanguage, pitch, stress, tempo, volume), leaving a conversation in a “social vacuum.”

Ried (1991) exposes the failings of CMC in this regard as follows: “Words, as we use them in speech, fail to express what they really mean once they are

deprived of the subtleties of speech and the non-verbal cues that we assume will accompany it. It is only the meanings of sentences that become problematic in computer-mediated communication. The standards of behaviour that are normally decided upon by verbal-users are not clearly indicated when information is purely textual.”

To take this argument further, Mackinnon (1995) states how CMC inadequacies overwhelm its success in comparison with offline world social structure. “Lacking physical reality, [CMC] users must create an explicit written language to convey meaning as well as emotion, physical qualities, and action. As a society based on language, it relies heavily on symbols, analogy, and metaphor to recreate or transfer physical matter and actions from the external world. But as these recreations are merely metaphors for, or “analogues” of, their physical counterparts, [CMC] can never be a mirror image of the external world.”

Another striking feature of networked CMC environments is the anonymity crisis. A user potentially enjoys opportunities to conceal his/her real offline identity in the online world. In comparison with other forms of communication, participants of CMC have substantially more control over their self-presentation. In face-to-face communication, each person involved in the communication process confronts the other, and obviously, they reveal their identities. In a telephonic mobile conversation, the speaker’s gender, to some extent their age, and emotional reactions like fear, anxiety, happiness, and tonal difference help to gauge varied qualities of semiotic reference to a speaker’s message framework.

Nevertheless, in the CMC, the textual form of message sharing bears a bare minimal imprint of social identities. The online message has the potential to confuse identity, which is counterproductive to those arguing that the content of messages reveals much about the nature of speakers (Lee, 1997).

The use of false identities, often of different sex, is widespread in electronic communities. Rheingold’s (1995) comparison of the offline and online world is an apt example. “The physical world is a place where the identity and position of the people you communicate with are well known, fixed, and highly visual. In cyberspace, everybody is in the dark. We can only exchange words with each other - no glances or shrugs or ironic smiles. Even the nuances of voice and intonation are stripped away. On top of the technology-imposed constraints, we who populate cyberspace deliberately experiment with fracturing traditional notions of identity by living as multiple simultaneous personae in different virtual neighbourhoods.”

The anonymity or the lack of clarity, single authorship condition has marred online-based research to a great extent. A researcher can never be sure of the demographics of site users, and according to the current level of technological improvements, there is no possible way to obtain identity confirmation. Most computer interfaces are either not designed to allow the user to question data validity or else are designed so anyone may change that data with a moderate level of technical skill. Until there is a new set of social

norms for validating computer information, there will be some uneasiness over the trust quotient of computer networks.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1. Explain the concept of time Shifting with suitable examples.

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2. Write about the pros and cons of cyberspace.

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3. Compare and contrast the face-to-face communication with online Internet as a Medium discussions.

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Activity – 1

Use any internet archives to visit the earlier generation USENET based online community and observe its past activities. Based on your observation, write a brief report [400 words].

1.3 TYPES OF INTERNET-BASED COMMUNICATION

The World Wide Web (WWW) has re-arranged the basic tenet of common person's communication activities in his/her day-to-day life. Even though

electronic communication is not an alternative to face-to-face (F2F) communication, Internet technology has considerably overcome the time and space barriers between two communicating points. Each fabric of social structure has established a meaningful connection with ICTs, ranging from e-business, online learning, e-governance, and e-shopping to online social movements.

The interconnection of multiple components of human society has increased information flow, cutting across national boundaries towards the creation of a “global village” predicted by Marshal McLuhan.

A multiplicity of forms and varieties of the communication processes can be carried out on the Internet with the fastest and cheapest ways and, more importantly, the widest reach. The telegraph detached the concept of transportation of communication from one point to another; the Internet went one step ahead and abolished the middle person to deliver messages directly and instantaneously. Moreover, it has facilitated the free exchange of information, unfettered, and unhindered by any censorship, which is another essential feature. The Internet has a profound impact on the day-to-day activities of the entire human social spectrum.

In the Internet medium, a user can explore a variety of possibilities independently. It is an opportunity, as explained by McLaughlin (1995), “To exchange electronic mail, transfer files, search databases, and retrieve information from remote libraries, take part in real-time conferences, run software on distance computers, and participate in discussion groups on varied topics.”

Computer-mediated communication is one type of communication facilitated by computer technologies. It is defined as “synchronous or asynchronous electronic mail and computer conferencing, by which senders encode in text messages that are relayed from sender’s computers to receivers.” There are two distinct types of CMCs existing over the computer technologies: synchronous and asynchronous, and the difference between the two CMCs is a temporal one. Synchronous CMC is produced when communication occurs simultaneously between two or more users, as in any normal telephonic or face-to-face or live video conferencing conversation. Asynchronous CMC is produced when communication is not simultaneous.

The most common and prevalent asynchronous CMC is electronic mail, in which a user can drop and receive a message at his/her convenience. Another popular category is social media postings (non-live messages), where the sender and receiver use a commonplace to read and post messages. In both cases, time is not a constraint for sending and reading the messages. On the other hand, in synchronous CMC, both the partners of the communication process must be available online. One-to-one or group text/audio/video-based chat/discussion via messenger software are popular forms of synchronous CMC.

Whether synchronous or asynchronous, “The CMC, it is claimed, will

1. create opportunities for education and learning;

2. create new opportunities for participatory democracy;
3. establish countercultures on an unprecedented scale;
4. ensnarl the already difficult legal matters concerning privacy, copyright, and ethics;
5. and restructure man and machine interaction” (Jones, 1998).

In the beginning, it was thought that CMC technology would overcome time and space barriers without having a centralised control authority to monitor the system. Besides, the universal spread of CMC has provided a facility to access an unlimited amount of data and reach other people. In the context of CMC’s capacity to create a global pattern of linkage, Lee (1997) elaborates its potential power as follows: “The information highway makes possible unprecedented forms of mediated communications. Never before have there been a means of communication, which have provided so many individuals with the ease and ability to engage in instantaneous, interactive communications with a broad and diverse public. The import of these new technologies, however, lies not simply in the novel forms of communication. What is also of profound significance is the ways in which these communicative possibilities lead to new types of identities and social relationships. Previously unimaginable, millions of individuals are increasingly interacting across time and space, and forming mutual bonds with others, most of whom they neither have met nor will ever meet face to-face.”

The Internet, due to its global reach and its rich multilingual context, has the potential to influence social relations. Unlike the traditional mass media, the online medium has an open architecture that restricts the efforts of legal authorities to regulate its activities. This, in turn, has facilitated greater freedom and space to its users and members to express their feelings on the net on issues that interest them. Thus, the Internet provides a technological infrastructure for CMC across both time and space, to create a group communication environment in which a form of virtual co-presence is established as a result of an individual’s online interactions. This has created a potential platform for virtual communities to flourish on the net. However, interconnected computers do not by themselves provide a congenial space for a group of people to float an online community in cyberspace; it is necessary to have sufficient human relationships.

1.3.1 Dynamics of Communication Process in CMC

Internet as a Medium Irrespective of its modality of presence, online or online overlapping with offline, any meaningful discussion in the CMC forum is determined by the quality of communication and its message. The very nature of CMC has abolished social and cultural domination in group dynamics. The computer medium inhibits users from transferring the social structures to CMC, and this inhibition is the result of the absence of, or limitations on physical proximity, face-to-face interaction, and non-verbal use.

People who are strong in articulation skills rather than social status play a dominant role in the text-based community. While analysing the social role of CMC technologies, Jones (1998) indicates two prominent usages: “Computers cut across or break down boundaries and break down hierarchies.” Similarly, Perrolle (1991) outlines the power and status structure of the communication process in the CMC milieu: “Computer mediated communication changes the nature of conversations between people. Research indicates that it alters the social norms governing conversation by removing elements of emotion and social control. It also provides the possibility of equal participation by obscuring the visual and verbal distinctions of status that gives high-ranking or aggressive people an advantage in face-to-face speech. Designs for cooperative work seek computer network support for circumstances in which opportunities for participation are enhanced, and opportunities for one speaker to control another are reduced. But computer-mediated communication can also embody inequalities in social relationships and can limit conversational participation.”

Factors like power and social status do not have any meaningful role in online activities. The virtual space in which people meet is neutral ground. Cyberspace belongs to everyone, yet no one. The conversation is the primary activity there, but it favours people good with their words rather than just fast with their mouths. The conversation is often much more thought out, and it is easier for everyone to express their opinion and not be dominated by one or two loud people. In an online community, every participant is in charge of his/her involvement. Online communities are entirely participant driven, and the conversation is about what a member has written.

A group of human beings settled in a new environment tend to formulate norms and values for the smooth conduct of the social processes of their new community. Likewise, in the online environment, virtual members also establish their own group-specific rules. Participants in CMC develop forms of expression that enable them to communicate social information and to create and codify group-specific meanings. Conformity with created norms also serves to socially negotiate group-specific identities, form relationships and organise interaction, and to maintain desirable social climates. In CMC, as in real life, relationships take time to build. The social information not available in the immediacy of face-to-face context can be gained verbally through computer-mediated interaction; the social penetration process just takes longer. People who meet online may then take that relationship offline if an opportunity arises.

There could be several reasons for a member to be associated with a virtual world. The loss of casual gathering places in our lives is one of the reasons as to why many people have turned to virtual spaces in which they can achieve some feeling of community. The virtual environment serves to fulfil a unique need for many people that they cannot find elsewhere in their lives. Although a multitude of special interest groups exists in the physical community, there is not always a forum located nearby to discuss the particular topic in which

each person is interested. Virtual platforms provide a forum for such discussions.

There is no topic under the sun left aside by these communities. Due to its worldwide reach, a group of enthusiasts joins together to stay abreast of the latest developments in their area of choice by having meaningful discussions on the digital forums. If the area of discussion is important and sensitive, in which a cross-section of human society has a sentimental touch, it corners massive attention from a relatively large number of members.

1.3.2 Cohesive Force of Online Group Communication

In the cyberspace area, people can roam freely and can create their virtual community that is far from the legal and nation-state purview. It comprises members distributed in all 24 time zones, who are equal in all possible social spheres in terms of shared belief, in the principles of free speech, individualism, equality, and open access.

Ananda Mitra (1997) applies theoretical concepts of offline community to online communities and states that the online community setup has a profound impact on the way we interact and the way we are going to interact with one another. Further, he adds that “In the electronic age, particularly in the age of the Internet, the organisation of human activities has become more complex with the availability of fast, efficient, and powerful means of communication that can have a significant impact on the way we organise the communities we live in and interact with. Moreover, that effect need not be restricted to specific geographic spaces but can be widespread as the tentacles of computer-mediated communication (CMC) reach across the globe.”

With this background information about the community and its set of rules that govern people’s interaction, it is clear that replication of existing offline cannons does not apply to virtual communities. At the same time, it is important to note that the emergence of online communities based on a different set of principles does not also occur in a social vacuum.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1. Differentiate the communication process between synchronous and asynchronous modes of online interactions.

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2. Differentiate the basic features of an online community with the Internet as a Medium offline community.

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Activity – 2

Enroll yourself in an online community that exists through any popular social media platform. Observe its activities for a fortnight, give your assessment - that should reflect your understanding of the theoretical concepts [400 words].

1.4 FORMS OF COMPUTER-MEDIATED COMMUNICATION

One of the classic studies in the area of CMC was done by Joseph Walther (1996, 2015), in which he classified three types of CMC –

- 1) Impersonal CMC;
- 2) Interpersonal CMC, and
- 3) Hyper-personal CMC.

These three forms of CMCs show the historical growth of research and understanding about the human relationship through computer-mediated interactions. The advent of technology has played an equally significant role in shaping and theorising human relationships through CMC.

1.4.1 Impersonal CMC

In the early stages of CMC, around the early 90s in the last century, available technologies mostly allowed rich-text based communications between people, like emails, text chat, and Usenet groups. Many early-stage research findings discussed the scenario of ‘scant social information’ and its implications in computer-mediated human communications in which nonverbal cues are minimal if not absent. However, many early-stage findings argued that in the impersonal CMC, the group tasks had better achievement rates due to lack of space in establishing social power structure in the textual CMC. However, in further research findings, the impersonal CMC concept was challenged in multiple dimensions.

1.4.2 Interpersonal CMC

The interpersonal CMC mainly equates the online discussion with the face to face communication settings. In interpersonal CMC, all efforts used by the participants of CMC to replicate the f2f settings, and various cues commonly adapted in interpersonal CMC (smileys are some of the adapted cues in interpersonal CMC). Since the efforts are employed to equate to f2f, normally interpersonal CMC requires more time to meet the expected outcome of communication settings.

1.4.3 Hyper-personal CMC

In the hyper-personal CMC environment, each stage of communication tools - sender, channel, receiver, and feedback - are employed carefully and optimally, more importantly, selective self-presentation in order to create an idealised perspective about the sender by the receiver. Since the communication between the participants is in a continuous process, sender channel-receiver-feedback is conducted continuously in order to maintain the idealised perspectives and impressions of each other.

1.5 VIRTUAL COMMUNITIES

An online or virtual community is an online entity in which a group of people gather together and exchange messages through digital networks. More importantly, these people gather into virtual communities based on their areas of common interest, and they may not necessarily come from the same geographical locations. Even many of the members might not have met each other in their life-time.

Harold Rheingold (1996), considered as the father of virtual community studies, defines virtual communities as “Cultural aggregations that emerge when enough people bump into each other often enough in cyberspace. It is a group of people who may or may not meet one another face-to-face, and who exchange words and ideas through the mediation of computer bulletin boards and networks. We do everything that people do when people get together, but we do it with words on computer screens leaving our bodies behind.”

A virtual community has its advantages compared to its counterpart in the offline world. CMC will do, by way of electronic pathways, what cement roads were unable to do - connect us rather than isolate us, put us at the controls of a “vehicle” and yet not detach us from the rest of the world. Unlike face-to-face interaction, in which relationships are initiated, and then topics of mutual interest sought, Internet users, can go directly to the topics that interest them and pursue interaction with like-minded others.

In furthering the concept of the virtual community, Rheingold puts forth the very purpose of online meetings. He proclaims it as “social aggregations that emerge from the [Internet] when enough people carry on those public discussions long enough, with sufficient human feeling, to form webs of personal relationships in cyberspace.”

Unlike the traditional mass communication setup, where central authority has control over the selection and production of messages consumed by millions, virtual communities offer a more democratic apparatus where access is broadly distributed and brings with it the option of interaction, offering new possibilities of community formation in the online platform. In the electronic communities, a set of shared practices help to produce the conditions that are similar to traditional communities outside of the realm of computers and virtual spaces defined by the “bit”-based technology of computers. Here, the community becomes central because the technology has now provided the ability to communicate across the boundaries and has removed limitations that the traditional community imposed.

Cyberspace technology’s support for the formation of virtual communities is a mere structural one, and it does not have any meaningful role in sustaining the group for an extended period. Human interaction and spirit of commonality is essentially the vital key to fostering virtual communities. The commonality that holds virtual community intact is the subject line criterion of togetherness, a feeling of connectedness that confers a sense of belonging.

Interconnected computers do not by themselves provide congenial space for a group of people to float an online community in cyberspace. Moreover, virtual communities of interest can be formed only if everyone adheres to a standard set of guidelines for organising the groups, subgroups, and topics. Internet as a Medium According to Baym (1995), the factors of temporal structure, external contexts, systems infrastructure, groups’ purposes, and participant and group characteristics are the most salient pre-existing forces in the development of the computer-mediated community.

Even though members of virtual communities come from diverse geographical points, they do share certain common traits. According to Lave and Wenger, participants of virtual communication can be classified into Peripheral (i.e. Lurker) – external, unstructured participation; Inbound (i.e. Novice) – a newcomer interested in the community and heading towards full participation; Insider (i.e. Regular) – fully committed community participant; Boundary (i.e. Leader) – a leader, sustains membership participation and brokers interactions; and Outbound (i.e. Elder) – in the process of leaving the community as a result of new relationships, new positions, new and outlook.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1. Explain the relevance of impersonal CMC in contemporary social media dominated online scenarios.

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2. Discuss the emergence of virtual communities on Facebook platform.

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Activity – 3

Visit any YouTube-based channel which has more than 1000 subscribers, analyse the type of participants it has. Categorise them as per points mentioned in the previous page.

1.6 LET US SUM UP

In this Unit, we have discussed the concepts related to cyberspace - an online platform that exists in digital networks through which we communicate our ideas, using various media elements such as text, photos, smiley, audio video content, etc. We have gone through the conceptual frameworks of cyberspace in which, unlike the offline world's social hierarchy, the knowledge matters in digital networks. Similarly, we have discussed about various formats of computer-mediated communication - synchronous and asynchronous, and different forms of CMC - impersonal, interpersonal, and hyper-personal. With the help of digital networks and through computer media communication, an online community emerges on its own by aggregating like-minded and common interest-oriented people gathered in cyberspace. This human settlement is a unique form of social relationship that exists within ourselves.

1.7 KEYWORDS

Asynchronous CMC: It is just opposite of synchronous CMC, yet another form of Computer-media communication.

This form of CMC happens with time constraints which means that the people involved in this communication process are not necessarily available on a real-time basis. The sender and receiver exchange information at different timings. There is a delay in feedback and responses. Email-based communication is one of the examples of asynchronous CMC.

Computer-Mediated Communication: A process of human communication of sharing words and ideas takes place between a group of people over digital networks. The information exchange may be on a real-time basis and delayed patterns, and the communication process uses multiple media elements.

Cyberspace: It is an electronic space or medium that exists in the networks of various networks. Cyberspace facilitates communication between human beings through computing devices and within/between computing devices. This conceptual electronic space is not constrained by distance and physical limitations.

Online/Virtual Community: There are many names associated with Online Community/Virtual Community/Digital Community/Web Community - but all refer to a group of people who aggregates themselves based on common interest, and they may come from any geographical place but use mostly asynchronous modes of computer-mediated communication. In the longer run, these communities tend to develop their social norms for their members.

Synchronous CMC: It is a form of Computer-mediated communication where the information exchange happens in real-time which means the people involved at both the ends of the communication process should be available in the computing networks but not necessarily in the same location. Virtual/video conferencing is one of the examples of synchronous CMC.

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1.9 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. There are two types of communication processes that are part of a digital platform - synchronous and asynchronous. Asynchronous communication does not require people to be involved in real-time while exchanging ideas and information. Interactivity is an important element of new media. It eliminates the time barrier for a user to access the contents conveniently anytime anywhere. The time-shifting facility enhances a user's involvement in the online environment, and hence, time is a crucial factor here. The virtual platform offers not only a multitude of topic areas but also the ability to participate at a convenient time. There is no weekly meeting to catch after work. Email is one of the popular examples of the asynchronous mode of computer-mediated communication.
2. Cyberspace is an electronic conceptual platform that exists in a network of networks. This platform facilitates communication exchanges between people, people, and digital devices and among the digital devices. The main advantage of cyberspace is that it cuts the physical distance between two communicating people, irrespective of their time zones. The quantum of the exchange of information will be huge and equally, and it might be in multiple modes of communication. The central negative dimension of cyberspace is that it reduces the exchange of non-verbal cues that might alter the length of the discussion as well as complete exchanges of ideas.
3. Face-to-face communication takes a topmost position in any ranking of communication settings. It is not comparable to any other alternative mode of personal interaction. However, cyberspace has provided a platform to interact with each other to overcome the physical constraints and different geographical locations. F2F is rich in cues and non-verbal communication, which has a strong influence on altering the course of communication between the persons involved. It is instantaneous, and the person involved in it takes a quick turn of his/her role. Whereas in computer-mediated communication, particularly in the synchronous CMC, there is a scope for instantaneous feedback and role change as

well, but the personal communication over cyberspace tends to lack non-verbal cues and signs, which might delay the information exchange.

Check Your Progress: 2

1. Synchronous based computer-mediated communication is being carried out on a real-time basis. The people involved in these activities, need to be present in the process in their respective places and be participants in the LIVE communication exchange. Whereas in asynchronous mode, as the name suggests, the communication process does not require the active presence of its participants.
2. An online community is a collection of people who hail from diverse geographical backgrounds. However, they join together in the online platform (be it a social media group member or email listserv group member) based on the common interest. In the online community, there are no physical or time constraints. In the case of an offline community, mostly it is location specified, and members usually come from the nearest places. However, the common interest could also be a binding factor here. Regular meetings or gatherings are the few options available to them to meet each other, and time is the biggest constraint here.

Check Your Progress: 3

1. Impersonal CMC refers to the communication exchanges that happen entirely in textual mode, which generally lacks non-verbal behavioural cues. These cues are crucial in routine face-to-face communication settings. Hence, impersonal CMC tends to be inferior to f2f settings. The scholar Joseph Walther gave critical inputs on impersonal CMC, however, later research findings gave critical assessments. In contemporary settings, the impersonal CMC is rarely noticed due to many advancements in the technology - audio-video conferences, smiley expressions, and more of multimedia oriented online information exchanges that gave ample scope to replicate nonverbal cues.
2. Virtual communities can exist in Facebook platforms - either the users are following an individual or a particular page or a group (it can be private as well as public; some of them are unmoderated while most of them are moderated). Being a member of this group or page or an individual page, they tend to exchange ideas with other members, over a period of time, the group or page or member of an individual develops its social norms. Thus, the emergence of a virtual community on the Facebook platform is possible.

UNIT 2 DIGITAL MEDIA & SOCIETY

Structure

- 2.0 Introduction
- 2.1 Learning Outcomes
- 2.2 Digital Media in Society
 - 2.2.1 Understanding Digital Media
 - 2.2.2 Evolution and Development of Digital Media
- 2.3 Concepts and Theories of Digital Media
- 2.4 Medium Specific Trends
- 2.5 Revolution within the Media Landscape
 - 2.5.1 Mass Media Adaptation into Digital Media
 - 2.5.2 Convergence in Digital Media
 - 2.5.3 Trends of Digital Media
 - 2.5.4 Revolution in Messaging
 - 2.5.5 Regional Languages in Online Media
- 2.6 Effects of Digital Media
 - 2.6.1 Cybercrime and Security
 - 2.6.2 Privacy and Surveillance
 - 2.6.3 Online Hate
 - 2.6.4 Disinformation and Misinformation
 - 2.6.5 Characteristics of Millennials
- 2.7 Let Us Sum Up
- 2.8 Keywords
- 2.9 References and Further Readings
- 2.10 Check Your Progress: Possible Answers

2.0 INTRODUCTION

Media as an umbrella term denotes ‘communication media’ and the institutions and organisations in which people work (the press, cinema, broadcasting, publishing, etc.) as well as the cultural and material products of those institutions. The denotations of media in the form of print and electronic media shifted from analogue (print and electronic) to digital media, encompassing the convergence and divergence of media. However, the transfer of electronic media to digital media has emulsified with the centrality of media, especially the new media. While television and radio sets have had a long presence in households in the developing world, new media has been omnipresent in the developed world: in living rooms, offices, and schools, in the streets, in playrooms and bedrooms. While in the developing world, the penetration of new media is novel but dynamic.

With the emergence of the Internet in the 1990s, the domain of media has restructured considerably, ranging from the point of media production, the processes through which information and representations of the media are distributed received, and consumed by its audience as well as how the media is regulated and controlled by the state or the market. However, digital media emerged as a concept in the 1980s with the shift from analogue to digital.

Nonetheless, the influx of digital media has also revolutionised the media playfield along with transforming the social, political, economic, and cultural spheres of life. Therefore, while studying digital media in society, we explore and scrutinise the new dynamics of digital media in the social, cultural, political, and economic realms, all of which constitute the society as a whole.

2.1 LEARNING OUTCOMES

After reading this Unit, you will be able to:

- describe the fundamentals of digital media in society;
- discuss the scope and growth of digital media in society;
- apply the theoretical understanding of digital media in real life;
- illustrate the trends, effects, and other issues that have emerged due to the advent of digital media;
- evaluate the development of digital media across the globe; and
- critically analyse the implications of digital media in society, politics, economy, and culture;

2.2 DIGITAL MEDIA IN SOCIETY

The materialisation of digital media has created a dual identity of the logic of media, prior to which media in a general sense consisted of print and the audio-visual medium. The legacy of media as a singular entity has disintegrated and re-integrated, through cultural, economic, and political processes, to pave the way for digital media. Therefore, the fulcrum of traditional media and new media is loosely edged on the aspects of production, consumption, and outlet patterns.

Media as an institution has become a space of converging and diverging spaces with overlapping characteristics of traditional media and new media, essentially due to the collapse of space and time as well as a mode of production. In the arena of news, newspapers, and magazines, television news channels, as well as online news platforms, have enriched the debate and the consumption patterns of news. This fact is evident across all stances of life and society.

To comprehend the extent of the influx of digital media in society, it will be pertinent to briefly trace the contextualisation of digital media along with the expanse of growth and development. We shall understand the characteristics of digital media to draw a comprehensive picture of digital media.

Furthermore, we shall take an overview of the manifestation of norms, theories, and concepts related to digital media in contemporary society.

2.2.1 Understanding Digital Media

Digital Media and Society the media sphere has continually been in a state of technological, institutional, and cultural change or development, from printing, photography through television to telecommunications. However, with the advent of digital media, the nature of change has not been the constant development that the media witnessed, but it increased rapidly from the late 1980s.

The bigger frames of cultural and social change that occurred at varying degrees from the 1960s onwards influenced the emergence of digital media, especially the shift from modernity to post modernity, intensifying the processes of globalisation as well as the shift from the industrial age of manufacturing by post-industrial information. This revolution introduced us to the concepts of digital media, online media, and new media.

New media, as a generic concept, introduces an arbitrary split between old and new media, inclusive of both digital and online media and other evolving media forms, implicitly emphasising the shift in media logic which delves into the openness and struggle between different ideas, users and logics.

Online media, as a concept, refers to the Internet, which is the prototypical new medium. The term online media prioritises the element of connectivity or how the connection is made with other media, mainly computers, and also more recently mobile telephones.

The concept of digital media arises from the characteristic that all information or data in these media is encoded in numbers, the most common being that of binary code of 0 and 1. Thus, technology is the defining dimension of the media which observes the shift of storing information digitally on a physical object, for example, USB flash drive, digital images, MP3, etc.

The ambiguous nature of new media as a term, which primarily leans on subjectivity, and the restricted perception of online media, which focuses on internet-based media only, leads us to focus on digital media in this Unit in convergence with the facet of online media since the Internet has enhanced the digital sphere.

In accordance with the focal point of this Unit, the characteristics of Digital Media can be categorised as follows:

- **Digital:** In a digital media process, all data are converted into numbers. Numerical representation makes digital information programmable, alterable, and subject to algorithmic manipulation. It can be compressed and decompressed using algorithms, allowing for large amounts of data to be stored and distributed efficiently.
- **Interactive:** According to Jensen (1998), interactivity is “a measure of the media’s potential and ability to let the user exert an influence on the

context and/or form of the mediated communication.” In this context, interactivity can be interpreted as a value-added characteristic of digital media.

- **Hypertext:** Hypertext is a form of text that is composed of nodes or blocks of text which form the content, the links between these blocks of content, and the buttons or tags that enact the link from one node to another. Thus, in a digital media process, hypertext combines traditional text with interactive branching to create a non-linear text MJM-028.indd 31 11-01-2021 12:10:44 32 New Media and Society which enables the ability to synthesise data and data retrieval at a later stage.
- **Dispersion:** Dispersion, here, refers to the diffusion of digital media, in comparison to mass media, at the level of consumption and production with the multiplication of sites and segmentation and resultant individualisation of media use.
- **Virtual:** The notion of virtuality emerges from the convergence of digital image technologies both with older kinds of analogue media and with computer-mediated telecommunications networks. The ‘virtual’ aspect of digital media focuses on the Internet and the World Wide Web; immersive, 3D, and spectacular image technologies; screen-based multimedia and animation.
- **Telepresence:** Digital media has the potential to alter our feeling of presence. Due to digital media, we gain the ability to simultaneously exist in two different environments at the same time: the physical environment in which our body is located and the conceptual or the interactional space where we are present with the use of the medium.

2.2.2 Evolution and Development of Digital Media

The emergence of digital coding can be attributed to Charles Babbage, who in the early 1800s, conceptualised codes and information by machines, followed by Ada Lovelace in 1822 and 1823, who wrote the first instructions for calculating numbers on Babbage’s machines.

However, the timeline of digital media commenced in 1941 with the invention of Z3, an electromechanical ‘Z’ machine by Konrad Zuse. It was the first working machine featuring binary arithmetic and a measure of programmability and is also regarded as one of the first computers, followed by the invention of Harvard mark 1, a large-scale electromechanical computer, in 1944. Digital devices such as the Xerox machine, communication satellite, microchip, virtual reality and augmented reality Head Mounted Display (HMD) system were developed after World War II, that is, 1947 and it went on till 1968.

In 1969, ARPANET was developed by the Defence Advanced Research Projects Agency of the US Department of Defence. It was the world’s first operational packet switching network. In 1970, Sony introduced the first videocassette followed by the invention of the computer floppy disks and microprocessors in 1971. The first video game console, the Magnavox

Odyssey, was launched in 1972, while Hewlett-Packard pioneered the Desktop computer in 1973.

Interestingly, in 1974, Nam June Paik, a 20th century South Korean-born US video artist, claimed to have coined the term ‘the information superhighway’, that is, a route or network for high-speed transfer of information. The year 1975 was a path-breaking because Microsoft was founded by Bill Gates, the first major ISP (Internet service provider) and the first hand-held mobile phone arrived. In 1979, the first cellular phone communication network started in Japan. The early years of the 1980s witnessed an upsurge in technological inventions and development such as compact disc players, Apple computers, CD Walkman, silicon microchip, CD-ROM, etc. to name a few.

In 1990, the possibility of the World Wide Web was described by Tim Berners-Lee, which further saw an upsurge in the extent of development in the media sphere. The following decade boomed with technological inventions across the globe with the first online website, digicam, short message service (SMS) to a mobile phone, DVD, digital still image camera, the first digital television service (DirectTV), etc. being developed.

In 1994, the US government released control of the Internet, and the World Wide Web (WWW) was brought in for public use which changed the media landscape considerably. E-commerce platforms (Amazon.com, Alibaba Group), online search engines (Google), email (Hotmail), social networking websites (MySpace) emerged in the latter half of the decade.

In 2004, the term ‘Web 2.0’ gained currency following the first O’Reilly Media Web 2.0 conference. Moreover, 2004 also witnessed the launch of the social networking website, Facebook followed by YouTube in 2005, Twitter in 2006, WhatsApp in 2009, LinkedIn and Instagram in 2010, Snapchat in 2011, and lastly, TikTok in 2012.

The digital media landscape is not just restricted to the devices or the networking websites as mentioned above as many more such digital media platforms emerged with a sudden boom which transformed the narrative of how we live and sustain in society.

2.3 CONCEPTS AND THEORIES OF DIGITAL MEDIA

The scope of concepts and theories of digital media is extensive. However, in this Unit, we are covering only the most central concepts and theories.

Marshall McLuhan: Remediation, Extension of Man, Medium is The Message

Here, we have outlined McLuhan’s ideas related to media and how it restructures in the context of digital media. We have concentrated on three key ideas: first, ‘remediation’, a concept that finds its roots in McLuhan’s view that ‘the content of any medium is always another medium’; second, his

idea that media and technologies are extensions of the human body and its senses; third, his view that 'the medium is the message'.

McLuhan's view of media as technological extensions of the body is his basis for conceiving four media cultures which are brought about by shifts from oral to written communication, from script to print, and from print to electronic media.

McLuhan configures technologies and mediums in this way because he views both as part of a larger class of things; as extensions of four of the five human senses: sight, hearing, touch, and smell. McLuhan also asserts that such extensions of our bodies, placed in the context of the body's whole range of senses (the sensorium), affect both our minds and our societies.

In the context of the medium is the message, McLuhan explains:

1. the power of media technologies to structure social arrangements and relationships, and
2. the mediating aesthetic properties of a media technology as our senses in different ways, the multidirectional simultaneity of sound as against a line of sight, the fixed, segmenting linearity of printed language, the high resolution of the film, or the low resolution of the TV, etc.

Raymond Williams: Social Shaping of Technology

In line with the 'social shaping of technology' school of thought, Raymond Williams stresses that the knowledge and acquired skills necessary to use a tool or machine are an integral part of any full concept of the technology.

Williams differentiates between:

- Technical inventions and techniques upon which technology depends, the alphabet, appropriate tools or machines for making marks, and suitable surfaces for accurately retaining marks;
- The substantive technology which, in terms of writing, is a distribution technology (it distributes language) and this requires a means or formscrolls of papyrus, portable manuscripts, mass-produced printed books, letters, or emails and other kinds of electronic text;
- The technology in social use. This includes (a) the specialised practice of writing which was initially restricted to minorities and then opened up, through education, to broader sections of society and (b) the social part of the distribution of the technologically reproduced language which again was only extended in response to perceived social needs.

Jean Baudrillard: The Real as Simulation

Jean Baudrillard argued that with globalisation and commodification, European and North American societies had changed their system of representation from one in which there was a clear separation between object and subject, between the real and its represented version, to a new state of reality as a simulation. In *Ecstasy of Communication*, Baudrillard states that

we have entered a new era of simulation or hyper-reality in which the screen and a network replace the scene and the mirror. To understand what he means by this radical replacement, we first need to understand what he means by the 'scene' and the 'mirror'. For Baudrillard, mirror and scene are both symbolic qualities of 'the object'. The object is what is external to or produced by the subject, the self, which corresponds to our intimate universe, our imaginary and symbolic world. In this equation, we are 'the subject'. Baudrillard argued that the opposition between object and subject, private and public used to define our relationship in and towards reality. Hence, we express our mental or psychological reality through making objects which open up the imaginary depths, the more profound scene of our life.

Actor-Network Theory

Actor-Network Theory denotes the relationship between humans, systems, and machines in computer networked communications. Actor-Network Theory (ANT) originated in the social sciences in studies of scientific, technological practices and networks. It is most associated with the work of Bruno Latour. Latour defines both human and non-human elements, structures in the environment as actors who can make other features dependent upon themselves. Actors have interests that can be aligned with different actors' interests to form an actor-network. Both humans and nonhumans may be 'actants' in a network made up of social groups, entities, and artefacts which then become enlisted to reinforce a position within network. ANT argues that nature and society are consequences, not causes Digital Media and Society of human, scientific, and technical work.

Post Modernity

The social and cultural dimension of globalised economic activity has been fruitfully defined as 'the condition of post modernity'. The condition of post modernity is another way of accounting for the interrelationship between technology, social, and cultural development. Post modernity is a period which deals with concepts that connect the emergence of new aspects of culture with changes in social life and the new globalised economic order. Postmodernism, distinct from the condition of post modernity, identifies trends, or movements in theoretical thinking, art, architecture, and cultural life. Postmodernism, in all its forms, argued that the changes in the world over, since the Second World War have been of such a magnitude that we can no longer continue with the rationalist modernist paradigm of thought and action. Postmodernists argue that a radical reorganisation of philosophical thought and cultural activity was needed and pointed towards cataclysmic events like the Second World War in Europe, the Holocaust, the collapse of the Soviet Union, the AIDs pandemic, and Chernobyl, which rationalist science and technological progress did nothing to stop.

Psychoanalysis

The concept of psychoanalysis, much like that of post modernity emerged with that of the advent of digital media. The critical paradigm of research encompasses psychoanalysis as an essential terminology which constitutes the essence of networks and digital media. Reflects upon the inner workings

of media and the mind. For understanding the concept of psychoanalysis, it is essential to know the importance of the work of Freud (1856-1939), as the notion of the self has been central to the content of media and artistic communication. Freud's radical contribution to the understanding of the human mind is based upon his elaboration of the concept of the unconscious. Prior to Freud, philosophy had equated the human mind with consciousness founded upon reason. Freud argued that only a small part of mental activity was conscious and that the unconscious consisted of inadmissible and involuntary ideas, which also motivate behaviour.

Freud developed a theory of the unconscious based upon his treatment of neurosis and his analysis of the content of dreams. Freud argued that the unconscious is made up of impulses, desires, or wishes, which get their energy from the physical instincts, of which sexuality was primary.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1. Describe the concept of digital media in your own words.

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2. What are the characteristics of digital media?

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3. What is the Actor-Network Theory? Describe in detail the contextualisation of digital media in Actor-Network-Theory.

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Digital media has influenced the emergence of trends that have completely transformed the dynamics of the media, political and social sphere such as the emergence of cyber laws, debates on net neutrality, the mobile revolution, digital literacy, and gaming.

Cyber Laws: With the arrival of digital media, policymakers across the globe have developed cyber laws for their respective countries, which govern the communication technology, particularly cyberspace or the Internet. Cyber laws and all the users of this space come under the ambit of these laws as it carries a kind of worldwide jurisdiction. Cyber law can also be described as a branch of law that deals with legal issues related to the use of inter-networked information technology.

In most of the developed countries, cyber laws have been developed stringently based on the norms of privacy and security of its citizens, especially the Western countries. In India, cyber laws are contained in the Information Technology Act, 2000 (IT Act) and subsequent amendments which came into force on October 17, 2000. The primary purpose of the Act is to provide legal recognition to electronic commerce and to facilitate the filing of electronic records with the government.

Net Neutrality: The concept of net neutrality focuses on the underlying assumption that all data on the Internet should be treated equally by corporations, such as internet service providers, and governments, regardless of content, user, platform, application, or device. Network neutrality requires all Internet service providers (ISPs) to provide the same level of data access and speed to all traffic. That traffic to any one or more service or website cannot be blocked or degraded. ISPs are not supposed to create unique arrangements with services or websites, in which companies providing them services are given improved network access or speed.

The term “network neutrality” was introduced in the year 2002. The concept was floated in response to efforts by the Federal Communications Commission (FCC), a United States regulator body, to require broadband providers to share their infrastructure with competing firms. Internationally, countries like the USA, Japan, Brazil, Chile, Norway, etc. have some form of law and order or regulatory framework in place that affects net neutrality.

In the Indian context, the Telecom Regulatory Authority of India (TRAI) adopted net neutrality regulations in July 2018. The net neutrality regulations in India affirms that internet access services should be governed by a principle that restricts any form of discrimination or interference in the treatment of content, including practices like blocking, degrading, slowing down, or granting preferential speeds or treatment to any content.

Mobile Revolution: Mobile media has firmly embedded itself into our Digital Media and Society lives in the contemporary world. Mobile phones have offered us the unique possibility of transcending the confines of space and location, allowing us to communicate with others regardless of where we

are. In doing so, they transform how we understand space, and have disrupted the boundaries we had placed between private and public space, and between working life and social-private life. With the infusion of mobile devices, the emerging highly mobile, personalised, atomised, hybrid work-social life has completely changed the dynamics of our way of life socially, culturally as well as politically.

In the political spectrum, horizontal social networks have developed extensively, however, the scope of democratisation of political mediation and the pressure of commercialisation and control of mobile media is still a contested issue. While in the socio-cultural outcomes of the mobile revolution, individual autonomy has been enhanced considerably with a renewed sense of creativity and use of imagination due to the influx of several messaging and calling digital platforms, however, the dominance of consumer culture has taken over the medium of face-to-face communication.

Mobile phones have also contributed to the exponential growth of diffusion with rates of more than 100% in some developed countries. Moreover, access to mobile phones in developing countries has increased substantially. The lower middle class is also increasingly dependent on mobile phones for information including access to the Internet on mobile phones, which in developing countries such as India is a significant source of information across all classes of the society.

Digital Literacy Initiatives: As technology transforms what and how we read and consume information, the notion of digital literacy becomes crucial. The American Library Association (ALA) defines digital literacy as “the ability to use information and communication technologies to find, evaluate, create, and communicate information, requiring both cognitive and technical skills.” In this context, digital literacy encompasses more than education because the Internet has become a key source of information. It is imperative to be digitally literate, to be able to comprehend what is authentic and reliable information while consuming information. Individuals who are aware of digital literacy understand the basics of Internet safety such as creating strong passwords, understanding and using privacy settings and know what or what not to share on social media.

While digital literacy as a concept has gained prominence across the globe at the institutional levels, India has embarked upon the National Digital Literacy Mission (NDLM) Programme, not just for students but for all strata of the society. The National Digital Literacy Mission Programme is a dynamic and integrated platform of digital literacy awareness, education, and capacity-building programmes that will help rural communities to take the lead in the global digital economy and help them in maintaining a level of competitiveness. It will also help in shaping a technologically empowered society. The vision of the Digital Literacy Mission (DLM) is to create multi-stakeholder, consortium and work with government and their various schemes and agendas to showcase in some of the panchayats constituencies that how making them digitally literate can bring about change in the scenario of governance, empowerment, social inclusion, educational approach, and employment.

Games and Gaming: The gaming industry in India is all set to grow with modern technologies like artificial intelligence. It is bound to revolutionise the gaming experience. To cite a study conducted by KPMG in the year 2019, the number of game development companies in India today stands at around 275. This number was a mere 25 in the year 2010. This number alone speaks volumes about the growth and possibilities of the gaming industry in India. With India having the world's largest youth population and second largest internet population, it makes the country one of the world's leading markets in the gaming sector. There are over 22.2 crore gamers in India who spend an average of 42 minutes per day on mobile games.

2.5 REVOLUTION WITHIN THE MEDIA LANDSCAPE

This sub-unit details out the revolution that has panned out within the media sector that contextualises the notion of digital media.

2.5.1 Mass Media Adaptation into Digital Media

The emergence of electronic media occurred in cohesion with the print media as the two forms of media complemented each other in terms of usability, access, and dissemination of information catering to different senses of an individual. However, with the arrival of digital media in the post-industrial era, digital media has completely altered the media landscape with digital media being a constant threat to print and electronic media due to its omnipresent and interactive features. This aspect is prominently dealt with within the recent KPMG report published in 2019. The overall industry performance in the digital media sphere has witnessed an increase of 38.5% from 2015-2019, whereas, in TV and print, films and radio, the industry performance is 9.9%, 5.6%, 9.6%, and 8.6% respectively. The digital segment is observed to be the torchbearer of the industry, and a similar outcome is projected up to 2024 in India.

Key technological innovations in the media and entertainment sector are disrupting the traditional approach to the dissemination of information, especially the smartphone user base, influx of mobile Internet, broadband connectivity in rural India, public internet access through the Common Service Centres network, e-governance, and embracing of digital consumption by both the private and public sectors of the economy.

The convenience of the digital media platforms and the ease of use and access has facilitated the disruption of mass media by the influx of digital media and is expected to dominate even further in the future.

2.5.2 Convergence in Digital Media

The first form of convergence to consider is technological convergence. Technological convergence is the movement of almost all media and information to digital electronic formats, storage, and transfer: the digitisation of all media, communications, texts, sound, images, and even

currency into a standard digital format or language. This process has involved a switch from analogue forms of media to digital forms.

Due to technological convergence, the governments or policymakers initiated the need to bring a change in the legislation to govern this new form of industry. This led to the second kind of convergence, here, referred Digital Media and Society to as regulatory convergence, which is a deregulatory strategy in the media and telecommunication industries adopted by the governments of many industrial economies since the mid-1990s. This set of regulatory changes has had a profound effect on the structure of the media, telecom, and computing industries and a significant impact on our current media culture.

Post the transformation of the regulatory framework, mergers, and acquisitions across the media industry was witnessed on a large scale, which was driven by a combination of technological developments and the overall climate of neoliberal deregulation and globalisation. The dominance of media conglomerates with large scale mergers and acquisitions paved the way for large scale cross-industry expansion or horizontal integration, in which a firm in one industry (for instance, telecommunications) expanded across to another industry (such as television broadcasting) and vertical integration, in which a firm that is concentrated on one point in the production chain of a sector (for instance, film production) expands into another part of the production chain in the same industry, such as film distribution. Both these types of expansion drove a general strategy of industrial convergence within the media and telecom sector, which is the third kind of convergence that occurred and changed the overall media landscape.

2.5.3 Trends of Digital Media

The most talked-about trend in digital media that is going to hit the Indian market is the rollout of 5G internet connectivity. Although India has predominantly been a 2G market since 2017, there has been a surge in the sale of smartphone devices owing to a combination of factors like falling data prices, better network coverage, low-cost 4G handsets, and the development of locally relevant content. The 5G network will not only allow the telecom sector to meet the requirement of high-speed data services, it will also support rich content-specific OTT services. With the increase in smartphone users to more than 800 million by 2022, an increasing number of Indians are likely to utilise digital distribution as the primary outlet for video consumption, thus emerging as a real threat to traditional, linear television in India.

The second most emergent trend of digital media can be that of 8K content and hardware. Leading TV manufacturers have started releasing 8K TVs in mature markets like the US, China, Japan, and Europe. With the advent of 5G, streaming of HD content is expected to become mainstream, which will benefit the sale of 8K TVs. In India, we are yet to see strong traction in the development of 8K content, but it is expected to get better in the coming years.

The third most relevant trend of digital media is that of Content Delivery Networks (CDN). CDN significantly reduces the site latency, boosts webpage load time, reduces bandwidth usage cost, and ensures global availability of content. AI-based predictive acceleration and the use of hyper-local CDNs are a couple of key trends in this arena. India is experiencing significant growth in data consumption-specifically concerning radio, hence, the importance of CDN is more significant than ever before.

The notion of digital labour is the fourth most important trend in digital media. Digital labour can be broadly classified into basic robotic process automation, enhanced process automation, and cognitive automation based on maturity levels. In the media and entertainment playfield, digital labour has found use in content generation, discovery and regulation, and also in support function automation. In India, several start-ups have come up focusing on intelligent automation, artificial intelligence, machine learning, and big data.

The fifth trend of digital media is that of Augmented and Virtual Reality which continues to disrupt the way media is created and consumed. Big players like Facebook, Google, Microsoft, and Magic Leap are coming up with innovative products and solutions to the market. In India, VR content production is gaining traction in areas like gaming, tourism, sports, advertising, etc. While usage of AR/VR is on the rise, widespread adoption is still in its early stages in India.

2.5.4 Revolution in Messaging

The speed, convenience, and versatility of social messaging have led to its integration within the entire customer journey, and even more significant than the traditional methods like emails, phone, or live chats.

WhatsApp and similar other social messaging apps have allowed people to share texts, videos, photos, and other content through a digital platform. Even though these apps are referred to as IP messaging apps, these have primarily become 'Social Networks'. The forerunners in the arena of IP messaging are WhatsApp, Viber, Google Duo, Hike, Skype, Facebook Messenger, VChat, Instagram Messenger, Snapchat, Telegram, etc. wherein WhatsApp takes the lead across the country.

A shift in the trend of messaging has emerged in the recent decade with voice calling taking a back seat and IP messaging becoming the predominant form of communication. Besides, these IP messaging platforms provide its users with the opportunity to place a video or voice call (also known as VOIP), thereby, changing the dynamics of the telecom sector in India and across the globe.

To keep up with the technological advancements and interests of the users and prospective users, IP messaging apps also delve into various forms of content sharing. The use of emojis, GIFs, emoji, stickers has become quite prevalent, especially amongst the younger audience along with the sharing of images, videos, documents, and location which have garnered worldwide attention as well.

2.5.5 Regional Languages in Online Media

At 97 million, the number of people speaking Bengali is more than the entire population of Germany. Similar comparisons made in the context of regional languages have revealed staggering numbers, for example, the number of people speaking Marathi (83 million), Telugu (81 million), and Tamil (69 million) is higher than the entire population of Turkey (82 million), France (65 million) and the UK (67 million) respectively.

The large audience size, combined with their preference to consume content in their preferred languages, has led to media players/platforms expanding their portfolios to offer dedicated regional language content on digital media platforms. Digital platform players such as Zee5, Voot, Hotstar, and Digital Media and Society Amazon Prime Video are investing heavily in original content creation in regional languages.

To attract the audience, OTT players are also tying up with major production houses in regional languages to purchase the rights of films, a trend which was already predominant in the General Entertainment channels (GEC) segment. In addition to this, players in the Indian music streaming industry like Gaana, Google Play Music, JioSaavn, Hungama, Spotify, Airtel Wynk, Apple Music, and Amazon Prime Music are gaining a foothold by supporting regional content, through partnerships with local brands like Saregama, T-Series, Zee Music, YRF, Sony Music as well as creating playlists considering the choices of Indian audiences and their preferences.

With the rise of web/voice searches in regional languages, search engines are facing a shortage of quality websites equipped to cater to these searches. This is giving regional marketers a unique opportunity to maximise impact by thriving on the relatively low competition for regional keywords. Brands are leveraging this take by optimising their content for these searchers. This gives them access to a focused target audience at rates which are much below the rates for English and Hindi; therefore, website localisation is at its peak in India in the present scenario.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1. What are the prominent trends specific to digital media in the Indian context?

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2. Describe the forms of convergence in digital media.

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3. Explain briefly how digital media has revolutionised the political and economic domains of society.

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2.6 EFFECTS OF DIGITAL MEDIA

The effects of digital media impact society at the micro as well as the macrolevels. Issues

about cybercrime, security, privacy, surveillance, online hate, disinformation, misinformation, and digital addiction are inherent when we discuss the effects of digital media.

2.6.1 Cybercrime and Security

Although, the advent of digital media has revolutionised the media playfield along with altering our personal and professional lives considerably, yet it has also let loose a set of crimes related to the digital forum, such as cyber attacks, identity theft, cyber theft, online scam, etc., affecting personal security of individuals as well as the national security of various nations.

While online scams, identity theft, and cyber theft pose a significant threat to individuals while trading identities on social media sites and e-commerce platforms, cyber warfare and national-level cyber-attacks have affected governance at a greater level.

Cyber intrusions and attacks have increased dramatically over the last decade, exposing sensitive personal and business information, disrupting critical operations, and imposing high costs on the economy. India became the second most affected country by cyber-attacks between 2016-2018. Cyber fraud and deception in the form of email scams, phishing, identity theft have become prominent in India as well some of the notable cases are the Union Bank of India Heist in July 2016, Wannacry Ransomware in May 2017, Data Theft at Zomato in May 2017, Petya Ransomware in June 2017. In 2018, the

Cosmos Bank Cyber-Attack in Pune, hacking of the Canara bank ATM, etc. were the major cyber-attacks of the year. In 2019, Pegasus Attack was one of the most prominent cyber-attacks which targeted Indian journalists, lawyers, and human rights activists.

Digital India Mission and increasing cyber security concerns have transformed this area into a multi-billion-dollar industry, currently valued at \$4.5 billion and expected to reach \$35 billion by 2030. In the present scenario, the three sectors which are heavily invested in cyber security efforts are the Government, Information and Technology Services, and Banking in India. In 2018, the Government of India came up with the National Cyber Security Policy, in consultation with all relevant stakeholders, user entities, and the public. The policy aims at facilitating the creation of a secure computing environment and enabling adequate trust and confidence in electronic transactions and also guiding stakeholders' actions for the protection of cyberspace.

2.6.2 Privacy and Surveillance

According to 2019, Forrester Global Map of Privacy Rights and Regulations: "Regulations that allow governments to access personal data of citizens are still undermining the overall privacy protections that certain countries offer their citizens."

India has been named as a country with minimal restrictions in terms of data privacy and protection where government surveillance is a matter of caution alongside countries with high-level government surveillance, such as China. According to industry experts, lack of constitutional provisions that enable monitoring of government activity could be one of the primary reasons for the high level of government surveillance.

2.6.3 Online Hate

The umbrella term of online hate encompasses cyber bullying hate speech, and online sexual harassment, which are highly dominant amongst adolescents and teenagers across the world.

Young people, who increasingly integrate many forms of social media into their intimate, social, and political lives, produce, are exposed to, and combat hate speech online. Further, they do so in a context where what constitutes hate speech, and what is recognised as racism, are critical dimensions of Digital Media and Society online engagement and discursive interaction.

Cyber bullying is a kind of harassment that uses electronic forms of contact. Bullying can include posting rumours about a person, threats, sexual comments, disclosing victim's personal information, or hate speech. Victims of cyber bullying show lower levels of self-esteem, increased suicidal tendencies, retaliation, and emotional breakdowns, and are also more prone to being frustrated, angry, or depressed. Many studies have shown that cyber bullying can be as harmful as traditional forms of bullying.

Anonymous chat rooms are popular among teenagers to make new friends and talk to different kinds of people. However, this modern twist to penpals comes with grave risks. Bullies hide behind the anonymity clause of websites to often target kids, either to make sexual and lewd comments or even send inappropriate content without their knowledge.

A need to combat online hate needs to be kept in mind by all the stakeholders involved, including the government, teenagers and parents, and appropriate steps taken towards the same.

2.6.4 Disinformation and Misinformation

The problems of disinformation, misinformation, and mal-information can be comprehended by the definitions of the terminologies given by United Nations:

- **Disinformation:** Information that is false and deliberately created to harm a person, social group, organisation, or country
- **Misinformation:** Information that is false but not created to cause harm.
- **Mal-information:** Information that is based on reality, used to inflict harm on a person, social group, organisation, or country. The rise in fake news and disinformation in India and across the globe has impacted the flow of information amongst the consumers, leading to the creation of bias and stereotypes. Digital media platforms such as WhatsApp, Facebook, Google, and Twitter are further fuelling the spread of disinformation and misinformation, thus, creating chaos and raising the issues of reliability and validity of the information. Social media and messaging apps are at the heart of the disinformation problems that India faces. According to a study conducted by Reuters Institute for the Study of Journalism at the University of Oxford, 52% of respondents say they get news via Facebook, and the same percentage say they get news via WhatsApp, which is owned by Facebook. With an estimated quarter-billion Indians having come online since the last general election in the year 2014, companies like Facebook, Google, and Twitter have become an integral part of the Indian media environment, including the disinformation problems that it faces. Indians are bombarded with fake news and divisive propaganda on a near-constant basis from a wide range of sources, from television news to global platforms like Facebook and WhatsApp.

Fake stories are spread by legions of online trolls and unsuspecting users, which has a dangerous impact. A rumour spread through social media about child kidnappers arriving in various parts of India has led to 33 deaths in 69 incidents of mob violence since 2017, according to IndiaSpend, a data journalism website.

Six months before the 2014 general elections in India, 62 people were killed in sectarian violence, and 50,000 were displaced from their homes in the northern state of Uttar Pradesh. Investigations by the police found that a fake video was shared on WhatsApp to whip up sectarian passions.

In 2018, “horrified by terrible acts of violence,” WhatsApp limited the number of chats that messages could be forwarded to in India from 256 users to five, and made it harder to forward images, audio clips, and videos (Some of these restrictions have since been rolled out worldwide).

2.6.5 Characteristics of Millennials

Millennials, or Generation Y (born in the 1980s and 1990s) and Generation Z (born in the 2000s) were the first generations to grow up with computers, the Internet, and smartphones as integral parts of their everyday lives. These “digital natives” spend an average of more than seven hours a day online, on their smartphones, or multiple devices at the same time (PC, laptop, tablet, and wearables).

The emerging digital media users are more active. While traditional media is consumed passively, consumers now have enhanced opportunities to share content, engage with content creators, participate in content, or even facilitate or sponsor content creation.

The millennial customer journey is highly influenced by technology, more so than any other generation. Millennials rely on having digital experiences with their banks than personal ones. Compared to older generations, millennials are most likely to use online- and mobile-banking channels due to easy-to-use websites and great apps. Technology equals convenience for millennials.

Investing in millennial engagement involves finding the right balance between traditional and digital experiences. As the world becomes increasingly more digital, opportunities to exceed millennial expectations and prepare brand marketing for Gen Z consumers will grow.

Traditional advertising tactics do not influence millennials. There is a new method of operation in response to millennials’ distaste for advertising as well as their increased social-media use and thirst for new technology. This powerful triad compels brands to keep pace with the latest trends in digital marketing. A shift from traditional media to digital media is a prerequisite, mainly social media websites and marketing on blogs and recommendation websites.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit. 1.

1. What are the effects of digital media on society in the contemporary world?

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2. Explain briefly the characteristics of millennials who are also Digital Media and Society considered digital natives.

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3. How has digital media enhanced the scope of regional languages in India?

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2.7 LET US SUM UP

In this Unit, we discussed the entrenchment of digital media in the society which has revolutionised not only our personal, cultural, political, but also our social spheres of life. We analysed and understood the difference between online media, new media, and digital media, along with discussing the characteristics of digital media such as digitality, interactivity, hyper textual, dispersal, virtuality, and telepresence. We traced the evolution and development of digital media enunciated with technological inventions and advancements, which led to the dynamics of digital media as we know today. We also explained a few related concepts and theories for contextualising digital media in society.

Apart from the overview of digital media, we outlined the trends specific to digital media- the emergence of cyber laws, net neutrality, the mobile revolution, digital literacy initiatives, and development in the gaming sector. Besides, the transformation within the spectrum of media due to the advent of digital media was examined. In this context, the realms of convergence and adaptation into digital media from traditional media were assessed, which overlooked the technological, regulatory, and industrial perspectives of change.

The revolution within the media playfield was also discussed along with the trends of digital media pertinent to India in tandem with governance, messaging, media corporates, artificial intelligence, and economy. Furthermore, the dimension of the effects of digital media across the ambit of society, politics, and economy was examined, primarily in the milieu of

cybercrime and security, privacy and surveillance, online hate, disinformation and misinformation, and digital addiction.

To sum up, it can be said that digital media has revolutionised the notion of media from the perspective of society as a whole, and it offers immense possibilities of expansion across all the spheres of life. However, the scope of digital media in society can be as adverse as it can be constructive. Therefore, it is imperative for the stakeholders involved in the domain of digital media to maintain check and balance for equilibrium in the impact of digital media.

2.8 KEYWORDS

Augmented Reality (AR): an enhanced version of reality created by the use of technology to add digital information on an image of something.

Digital Economy: an economy that focuses on digital technologies, i.e. it is based on digital and computing technologies.

Digital Media: digitised content that can be transmitted over the internet or computer networks.

e-Governance: application of Information and Communication Technologies (ICTs) for delivering government services through the integration of various stand-alone systems between Government-to-Citizens (G2C), Government to-Business (G2B), and Government-to-Government(G2G) services.

Over-the-Top (OTT) Media: film and television content provided via a high-speed Internet connection rather than a cable or satellite provider.

Virtual Reality (VR): the use of computer technology to create a simulated environment.

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2.10 CHECK YOUR PROGRESS: POSSIBLE ANSWER

Check Your Progress: 1

1. The concept of digital media arises from the characteristic that all information or data

in these media is encoded in numbers, the most common being that of binary code of 0 and 1. Thus, technology is the defining dimension of the media which observes the shift of storing information digitally on a physical object, for example, USB flash drive, digital images, MP3, etc.

2. Characteristics of Digital Media:

- Digital
- Interactive
- Hypertext
- Dispersal
- Virtual
- Telepresence

3. Actor-Network Theory denotes the relationship between humans, systems, and machines in computer networked communications. Actor-Network Theory (ANT) originated in the social sciences in studies of scientific and technological practices and networks and is mostly associated with the work of Bruno Latour. Latour defines both human and non-human elements and structures in the environment as actors who can make other features dependent upon themselves. Actors have interests that can come into alignment with different actors' interests to form an actor-network. Both humans and nonhumans may be 'actants' in a network made up of social groups, entities, and artefacts which then become enlisted to reinforce a position within a network. ANT argues that nature and society are consequences, not causes of human scientific and technical work.

Check Your Progress: 2

1. Prominent trends of digital media in India:

- The emergence of Cyber Law
- Net Neutrality
- Mobile Revolution
- Digital Literacy Initiative
- Gaming

2. Forms of convergence in digital media:

- Technological convergence
 - Regulatory convergence
 - Industrial convergence
3. Digital media revolutionised the political domain: e-Governance, regulatory convergence, cyber laws. Digital media revolutionised the economic domain: digital media corporates, digital economy, industrial convergence.

Check Your Progress: 3

1. Effects of digital media:
 - Cybercrime and security
 - Privacy and surveillance
 - Online hate
 - Disinformation and Misinformation
 - Digital Addiction
2. Characteristics of Millennials:
 - Digital natives
 - Active users of digital media
 - The convenience of digital media for banking
 - Share content, engage with content creators, and participate in content creation
 - Digital marketing and brand marketing for millennials
 - Social media websites and marketing for recommendations
3. Scope of regional languages in digital media:
 - Regional language content on digital media platforms
 - Original content creation on OTT platforms
 - OTT platforms tying up with major production houses for rights of films
 - Regional languages on digital music streaming industry
 - Digital marketing in regional languages gaining prominence
 - Localisation of websites

UNIT 3 ISSUES OF ACCESS AND PARTICIPATION

Structure

- 3.0 Introduction
- 3.1 Learning Outcomes
- 3.2 Digital (In) Equality: Conceptual Framework
- 3.3 Evolution and Development of ICT
 - 3.3.1 Evolution of ICT
 - 3.3.2 ICT Technologies
 - 3.3.3 Developmental Role of ICT
- 3.4 Growth and Diffusion of ICT
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- 3.5 Digital Divide
 - 3.5.1 Determinants of Digital Divide in India
 - 3.5.2 Implications of Digital Divide
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3.0 INTRODUCTION

The digital media system has been in existence for the past 25 years in India. We have experienced these digital media as luxury gadgets and services in the initial phases, but later they became a part of life for many, if not for all. While there were more concerns about the physical access to Information and Communication Technology (ICT) in the initial stage, later the divide between 'haves' and 'have-nots' has significantly changed into digital (in)equality as the access and usage are increasing which is becoming a significant issue of concern.

The developed countries have reached a saturation point in terms of the penetration of ICT, however, the developing countries are witnessing vast inequalities within the nation. Most urban centres are highly saturated compared to semi-urban and rural areas. Interestingly, higher growth rates of penetration, access, and usage are being observed in semi-urban and rural areas.

There is a need to pay close attention and scrutiny to the access and participation of users of digital media in the Indian context. Digital inequalities are major concerns beyond physical access. There is a need for quality internet connection, availability of regional language content and its

relevance, opportunities to utilise the ICT tools for routine work, and more importantly the skills and competencies of users to interact with the various ICT tools in order to maximise the benefits of digital media.

After reading this Unit, you will be able to learn about the growth and diffusion of Information and Communication Technology (ICT) in India. This Unit seeks to develop an understanding of the concepts, definitions, nature, roles, functions, diffusion, and access to ICT. The term ‘ICT’ is used here in a broader sense and emphasises how ICT tools -- fixed-line phone, mobile-cellular phone, and the Internet have diffused and is being accessed across the country. The Unit also discusses how the digital divide still exists between rural and urban areas, and across the states. The key initiatives taken by both governments and private organisations to bridge the digital divide in the country has also been touched upon.

3.1 LEARNING OUTCOMES

After reading this Unit, you will be able to:

- understand the conceptual frameworks of digital inequalities;
- describe the evolution, nature, and the developmental role of ICT;
- trace the growth and diffusion of ICT in the global context;
- discuss the determinants of digital divide; and
- describe the impact of the digital divide and initiatives to bridge it in the Indian context.

3.2 DIGITAL (IN) EQUALITY - CONCEPTUAL FRAMEWORKS

Digital inequality is the latest connotative term associated with an earlier concept called the digital divide, which explains that there is uneven distribution, access, and participation of digital infrastructure in the society. There are two concepts linked together in ‘digital inequality’ — digital refers to the technological resources pertinent to Information and Communication Technology (ICT), and inequality denotes the unevenness in accessing and participating with the ICT resources. In a related context, we can refer to one of the frequently quoted definitions of the Organization for Economic Co-operation and Development (OECD) on the digital divide in order to understand its relevance in today’s context. OECD (2003) defines the digital divide as “The gap between individuals, households, businesses, and geographic areas at different socio-economic levels with regard to both their opportunities to access information and communication technologies and their use of the Internet for a wide variety of activities.”

According to Van Dijk (2017), the digital divide needs to have much more clarity. Dijk says, “In fact, it is a metaphor that has inspired at least four misunderstandings. First, the metaphor suggests a simple distinction between two divided groups with a yawning gap between them. Second, it suggests that this gap is difficult to bridge. Third, it can imply absolute inequalities

between those who are included and those who are excluded, whereas inequalities are of a more relative kind. Finally, the digital divide is not a static and permanent condition.”

As we discussed in the previous Unit on the status of digital media penetration in India, according to TRAI 2020 data, we have nearly 51% MJM-028.indd 50 11-01-2021 12:10:44 51 Issues of Access and Participation of the Internet and 87% of mobile penetration. Nevertheless, if you dissect these percentages further, you will find significant disparities between locations of residence like urban or rural, or on a gender basis, at the level of Internet connections, on the types of computing/mobile devices used and its quality, on the availability of regional language content and more importantly users’ skills and competencies in using digital devices and particularly the relevance of these devices for personal usages.

However, three essential dimensions need to be considered while assessing digital inequality — access to networks, competencies to use devices and relevance as well as usage of digital platforms. Digital inequality is not about the physical presence or absence of network platforms. However, beyond that, several factors are determining the quality of access, competencies, and relevance, like economic conditions, geographical locations, educational background, age, family background, availability of regional content, occupation, literacy, and technical skill sets, etc. Hence, digital inequality is a socio-economic issue, as those who have better access to ICT tend to get more benefits. On the other hand, groups that lack better access and necessary digital competencies and skills tend to spend more time with entertainment content, as they lack the skills to use ICT tools for their professional or political usage. In the global context, the developed countries are in a better position to implement newer ICT tools quickly due to their political structure, economic conditions, market structure, and better investment in research and development (R&D) sectors. Many developing countries fail to meet if not all, many of these factors, and hence it affects them to roll out the latest technology to boost communication infrastructure. Huang and Chen (2010) observed that “There is no universal trend — the severity of the divide decreases in some cases, but it increases in others.”

Mossberger, Tolbert, and Stansbury (2003) have reformulated the digital divide further into four-forms of divides - the access divide, skills divide, economic opportunity divide, and democratic divide. According to them, “The access divide includes the location of computer and internet use and the frequency of internet users in each location. The skill divide is measured in terms of technical competencies and digital literacy. The economic opportunity divides are conceptualised as experiences, beliefs, and attitudes regarding computers and economic advancement. Finally, the democratic divide is defined as attitudes and experiences regarding internet use for political activities.”

Schweitzer (2015) further categorised the digital divide into many levels, — “The digital divide encompasses differences in both access (first-level digital divide) and usage (second-level digital divide) of computers and the Internet between (1) industrialised and developing countries (global divide), (2)

various socio-economic groups within single nation-states (social divide), and (3) different kinds of users with regard to their political engagement on the Internet (democratic divide).”

If you keenly observe the historical developments of the emergence of digital inequalities since the 1990s, you will notice that in the earlier period (1995-2000), the majority of the discussion happened around physical access to ICT infrastructure, between 2000-2005 the competencies and skills dominated the discussion, and in last one decade, the focus area is being shifted to relevance and usage of ICT for the public as well as private purposes. Yet another set of divisions happened in the discourse of digital inequality over the period — horizontal versus vertical perspectives of digital inequalities. The horizontal perspective delves into the quantitative dimension of digital inequalities, such as the quantum of ICT infrastructure and its affordability and accessibility at the individual, national and international levels. Whereas in a vertical perspective, the qualitative dimension of digital inequalities is interpreted in terms of competencies, skills, usages, and outcome of its usage.

Digital inequality or digital divide is being extensively scrutinised from diverse basic as well as applied research perspectives over the past two decades. Access to communication technology and usage of ICT tools are mostly studied from the media studies discipline, and perception and motivations are focal points from psychology perspectives, inequality, and its related issues from the sociological point of view and media information literacy from pedagogical perspectives.

Van Dijk quotes the theoretical evolution of academic works related to digital inequalities that give a broader framework for understanding this crucial area from the research point of view. He says, “In social and communication science, these stages of access are dealt with by technology acceptance theories. Most psychological theories such as the technology acceptance model and the theory of planned behaviour deal with access focusing on motivation and attitudes. Examples are perceived usefulness, ease of use, and subjective norms that affect behavioural intention to gain access to digital media. As soon as acceptance reaches the stage of decision making, adoption theories such as diffusion of innovation are used. These are theories more rooted in sociology and communication and focus on social and contextual factors. In the post-adoption stages (initial and continuous use), social, cultural, and communication theories figure examples include domestication theory, uses and gratification theory, and social cognitive theory, leading to the model of media attendance.”

Van Dijk (2017) has proposed a model of the resources and appropriation theory. The main aspects of this theory are:

- 1) Categorical inequalities in society produce an unequal distribution of resources.
- 2) An unequal distribution of resources causes unequal access to digital technologies.

- 3) Unequal access to digital technologies also depends on the characteristics of these technologies. Unequal access to digital technologies brings about unequal participation in society.
- 4) Unequal participation in society reinforces categorical inequalities and unequal distributions of resources.

With the above theoretical understanding of digital inequalities, let us examine the ICT access and participation from the Indian perspective.

3.3 EVOLUTION AND DEVELOPMENT OF ICT

All of us would agree that we cannot imagine living in a modern society without using Information and Communication Technology (ICT). Most of us use the Internet, mobile phones, computers, and smartphones, and these ICT tools have become an integral part of our daily lives. They have also integrated into every aspect of our business economic activities.

3.3.1 Evolution of ICT

The information age began in the mid-nineteen century after the industrial age when the work process starts involving the creation, distribution, and application of information. It has evolved in phases from a society having radio as the primary source of information to television as a popular source of information and entertainment. Later, we moved to the wireless age or digital age of mobile phones, personal computers, and the Internet. Over the years, these changes in technologies have dramatically transformed the sources and flow of information. The sizes of technological tools have reduced drastically, and the speed to operate and process data and communication of information has also increased manifold. We can identify the following five phases of ICT evolution.

The first phase of ICT evolution started with the creation of electromagnetic calculators during the Second World War (1939-45) followed by the invention of transistors in 1947, which led to the development of smaller but handy computers. With the changing technologies of input and output devices from punch cards to magnetic tapes, “Computer” gained popularity.

The second phase of ICT evolution is associated with the development of personal computers during the 1970s. The development of chip technology and manufacturing magnetic discs transformed huge computers into desktops, which are called personal computers. Word processing, accounting, and graphics became accessible.

The third phase of ICT evolution is related to the development of microprocessors. A microprocessor, a multipurpose, programmable device that accepts digital data as input, processes it according to instructions stored in its memory and provides results as output. Microprocessors reduced the cost of processing power.

The fourth phase of ICT evolution is around networking. Starting from connecting computers within a small geographic area for use in the field of Defence and Education to connecting the computers worldwide, resulting on the Internet is the landmark development in the field of ICT. This resulted in breaking the distance barrier across the geographies, and the world is called a global village. Social networking is a by-product of this worldwide connection. The introduction of broadband has accelerated this evolution.

The fifth phase of ICT evolution is related to the development of wireless, which refers to the present phase. This phase began with the invention of the mobile phone. The initial mobile phones were large and bulky. A greatly expanded range of functions has accompanied a reduction in size with the advent of smartphones. Now mobile phones, particularly smartphones, are used for talking, transmitting messages, pictures, and music, browsing, navigating, etc.

3.3.2 ICT Technologies

There is a range of technologies that fall under the category of ICT— from traditional to modern communication systems. Some of the standard information and communication technologies are given in Table 3.1, which presents ICT technologies and their roles or functions.

Table 3.1: Technologies and Functions of Information and Communication Technologies

ICT Technologies	Roles/Functions (Information)
Personal Computers, Digital camera, Scanner, Smartphone	Creation
Calculator, PC, Smartphone	Processing
CD, DVD, Pen Drive, Microchip, Cloud	Storage
PC, TV, Projector, Smartphone	Display
Internet Teleconference, Video conferencing, Mobile technology, Radio	Transmission
E-mail, Mobile-cellular phone	Exchange

3.3.3 Developmental Role of ICT

Information and Communications Technologies (ICT) play an important role in the development process as it contributes towards national income or GDP and improves people's living standards by facilitating access to more livelihood opportunities, and global information. It promises a fundamental change in all aspects of our lives, including knowledge dissemination, social interaction, economic and business practices, political engagement, media,

education, health, leisure, and entertainment. Some of the vital developmental roles of ICT are summarised in Table 3.2.

Table 3.2: Developmental Role with ICT tools

Development Role	ICT-Tools
<i>Improved life</i> • Better health services • Better education services • Better economic opportunities	mobile phone, telecenters, health information systems
<i>Improved government services</i> • Efficient services to citizens • Visible direct services • Better planning and implementation of infrastructure	e-government services, geographical information systems
<i>Enhanced internal economic activities</i> • Better financial services to citizens and business • More efficient enterprises • Improved efficiency in the vital sector of agriculture	Internet & mobile banking, e-commerce; agriculture information for marketing, weather, and others
<i>Improved civil society</i> • Increased efficiency • Reduced corruption • Improved social and political participation of poor and disadvantaged groups (women, slums, remote/rural), • Information as a means to effect negative attitudes such as apathy and indifference	Records on land reform Local information through ICTs e-services applications to provide broader information on welfare schemes to civil society Transparent information access through the Internet, websites, applications or mobile messages, etc.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1. Explain the characteristics of digital inequalities?

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2. Trace the historical development of ICT?

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3. Describe three broad applications for ICT technologies?

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3.4 GROWTH AND DIFFUSION OF ICT

The impact of ICT can be seen within the perspectives of two interrelated aspects: diffusion and growth. While growth refers to the expansion of ICT such as the number of mobile phones and Internet subscribers, diffusion refers to the distribution of ICTs use across people from different places like rural-urban, developed-underdeveloped regions, rich-poor people, and other disadvantaged people—women, handicapped, socially included—SC and STs, etc. The diffusion part affects human welfare such as education, health, socio-economic development of people, economic growth, increased productivity, and competitiveness in different sectors of the economy. In this section, we are focusing on the growth and diffusion of ICTs technologies, particularly landline phones, mobile phones, and the Internet.

3.4.1 Growth of ICTs

Fixed-Phone and Mobile-Cellular Phone

We can see from Table 3.3 that the fixed-line phone subscribers were at a peak in the early 2000s and started declining after it, while mobile phone penetration gained movement during the same period. The fixed-line phone subscribers globally declined (1243 million in 2005 to 1004 million in 2016), on the other hand, the cellular phone subscribers increased more than three times during the same period (2205 million in 2005 to 7511 million in 2016). The growth of cellular phone subscribers is significantly higher in developing countries (1213 million in 2005 to 5909 million in 2016) compared to the developed world, (992 million in 2005 to 1603 in 2016). This is because building telephone infrastructure was difficult in remote and rural areas in most of the developing countries like India, which has adopted cellular phones in more significant numbers. The majority of the fixed phone (41 per

cent) and mobile-cellular phone (54 per cent) subscribers reside in Asia & the Pacific, but the penetration rates are low compared to other parts of the world.

Table 3.3: Growth of Fixed Line Phone and Mobile Phone Subscribers (Tele density)

	Millions				Per 100 inhabitants			
	2005	2010	2015	2016	2005	2010	2015	2016
Fixed-Phone								
Developed	570	553	490	479	47.2	44.6	39.0	38.1
Developing	673	676	556	524	12.7	11.9	9.2	8.5
World	1243	1229	1046	1004	19.1	17.8	14.3	13.6
Asia & Pacific	559	553	439	410	15.1	14.2	10.8	10.0
Mobile-Cellular								
Developed	992	1404	1563	1603	82.1	113.3	124.5	127.3
Developing	1213	3887	5621	5909	22.9	68.5	92.7	96.3
World	2205	5290	7184	7511	33.9	76.6	98.2	101.5
Asia & Pacific	833	2614	3778	4070	22.6	67.3	92.6	98.9

Source: ITU, 2018

We can further see from Table 3.3, the penetration rate of the fixed-line phone and mobile-cellular phone as is measured by tele-density (phone per 100 people). Similar to the subscribers, the tele-density of fixed-line phones declined from 19.1 per cent in 2005 to 13.6 per cent in 2016. Similar trends were also observed both in developed, developing countries and Asia & Pacific. On the other hand, the teledensity of mobile phones increased almost three times, from 33.9 per cent to 101.5 per cent during the same period. In particular, the developing countries (22.9 per cent in 2005 to 96.3 per cent in 2016) have benefited most by mobile phone (82.1 per cent in 2005 to 127.2 per cent in 2016) with a higher rise of its subscribers compared to developed countries, which had already reached 82 per cent tele-density in 2005.

India's telecom (mobile phone and fixed services) subscribers increased from 0.1million in 1947 to 5 million in 1991 and 29 million in 2000 to 1183.5 million by 2019 (Table 4). It is the second-biggest mobile market in the world in terms of subscribers, second only to China. Indian telecom subscribers account for 28 per cent of Asia Pacific's total telecom subscribers and 15 per cent of the global subscriber base (ITU, 2018). Mobile phones

have mainly contributed to the phenomenal growth of phone services during the last decade, which crossed the fixed-line phone subscribers in 2004. Presently the mobile phone segment (1161.8 million) dwarfs the fixed segment (21.7 million) by a ratio of 53:1. The mobile phone subscribers constitute 98 per cent and have increased by almost 16 times between 2005 and 2019. Today India is ranked second in the world in terms of mobile phone subscribers (India is ranked second after China (TRAI, 2018)), which is attributed to various factors such as low prices of services, rise in people's income, competition, and the introduction of micro prepaid connection and low handset prices (Guha & Kaushal, 2010). Mobile phone subscriber annual growth rate has fluctuated from high at 13 per cent in 2010-11 to negative (-7 per cent) in 2012-13 and only 1 per cent in 2017-18 and witnessed negative growth during 2018-19 period.

Table 3.4: Telecom Subscribers (in millions) in India, 2019

Year	Fixed Phone	Mobile-Phone	Total
2000	27	2	29
2005	49	76	124
2010	35	752	787
2011	34	866	900
2012	32	929	961
2013	30	868	898
2014	29	905	933
2015	27	970	996
2016	25	1034	1059
2017	24	1170	1195
2018	23	1183	1206
2019	21.7	1161.8	1183.5

Source: TRAI, 2019

However, this fluctuation in growth was mainly due to the removal of inactive subscribers by the operators to improve their operational performance and compliance with subscriber verification norms. TRAI introduced registration of active users via Visitor Location Register (VLR), and the estimated number was around 50 per cent of the total subscribers in 2011. *[The regulator measures the number of active subscribers via its Visitor Location Register (VLR), a temporary database of subscribers who have roamed in a particular area. Exactly one VLR serves each base station*

in the network; hence a subscriber cannot be present at more than one VLR at a time. VLR data is calculated based on the date of Peak VLR of the particular month and gauged from the switches having a purge time of not more than 72 hours (TRAI, 2018).]

This is due to the overestimation of subscribers, inactive connections, and usage of multiple Subscriber Identity Module (SIM) cards. About 35 per cent of mobile connections were inactive, and 15 per cent had multiple SIMs either on the same phone or in different phones. Applying filters of inactive users and multiple SIM holders, it translates that about only half of the total subscribers were active in 2011. Over the years, after shedding off inactive subscribers by the operators, still, 16 per cent of the total subscribers were inactive in March 2018. The decline in mobile phone subscribers in the recent period was the consolidation of the market after the entry of **Reliance Jio**, several small operators have either merged or exited from the market.

Internet-Broadband

Internet penetration has picked in recent years mainly after the entry of wireless Internet. Wireline Internet subscribers increased more than four times, from 220 million in 2005 to 917 million in 2016 globally (Table 5). The rate of growth is significantly higher in developing countries (71 million to 535 million) than in the developed countries (148 million to 382 million) during the same period. The active wireless Internet subscribers increased rapidly in recent years, from 807 million in 2010 to 3864 million in 2016 in MJM-028.indd 57 11-01-2021 12:10:45 58 New Media and Society the world. The pace of active wireless Internet subscribers' growth is also higher in developing countries compared to that of the developed world. However, the overall penetration rate of wireline Internet (12.4 per cent) and wireless Internet (52.2 per cent) was far less in developing countries (8.7 per cent and 43.6 per cent) compared to developed ones (30.3 per cent and 94.4 per cent) in 2016. The Asia & Pacific region also shows a low penetration rate of both wireless Internet (47.4 per cent) and wireline internet (11.3 per cent).

Table 3.5: Internet-Broadband (Subscribers)

	(Millions)				Per 100 inhabitants (Tele-density)			
	2005	2010	2015	2016	2005	2010	2015	2016
Active Wireless- Internet subscriptions								
Developed	NA	554	1118	1189	NA	44.7	89.1	94.4
Developing	NA	253	2179	2676	NA	4.5	36.0	43.6
World	NA	807	3297	3864	NA	11.5	45.1	52.2

Asia & Pacific	NA	286	1554	1951	NA	7.4	38.1	47.4
Wireline Internet subscriptions								
Developed	148	291	370	382	12.3	23.5	29.5	30.3
Developing	71	236	472	535	1.3	4.2	7.8	8.7
World	220	526	842	917	3.4	7.6	11.5	12.4
Asia & Pacific	80	214	408	463	2.2	5.5	10.0	11.3

NA=Not Available; Source: ITU, 2018

In India, the Internet was started first by Videsh Sanchar Nigam Limited (VSNL) in 1995 via dial-up services in six cities. Moreover, the pace of internet penetration was very slow at only 0.1 per cent in 1999 due to the high price of the services and low penetration of landline phones. The Internet penetration rate picked up only after an increase in the number of Internet service providers in 2006 and the availability of wireless Internet networks and smartphones in recent years. Currently, India has over 493 million Internet subscribers, making it the second-largest Internet user in the world, ranked only behind China. Over the last few years, an increase in Internet penetration has been driven mainly by wireless Internet with the availability of cheaper smartphones.

The number of Internet subscribers in the country increased at an annual growth rate of 43 per cent, with the number reaching 604 million in 2019 from 5.6 million in 2000 and 92.5 million in 2010 (Table 3.6). This growth is attributed to wireless Internet with cheaper smartphones, faster connectivity, and affordable services. Wireless Internet subscribers in the country witnessed rapid growth and reached at 604 million compared to 21 million wired Internet subscribers. This indicates that the wireless Internet is mainly responsible for rapid internet penetration growth in recent years.

Table 3.6: Internet Subscribers (in a million) and Penetration (in percentage)

Year	Internet subscribers	Penetration (%)	Total Indian Population
1994	0	0.00	945
1995	0.29	0.03	963
1996	0.49	0.05	982
1997	0.7	0.07	1000
1998	1.43	0.14	1019
1999	2.8	0.27	1038

2000	5.59	0.53	1056
2005	27.4	2.39	1147
2010	92.55	7.5	1234
2011	125.88	10.07	1250
2012	159.14	12.58	1265
2013	193.28	15.1	1280
2014	233.1	18	1295
2015	353.7	27	1310
2016	422	31.87	1324
2017	445.96	33.33	1338
2018	493.96	36.54	1352
2019	604.21	44.23	1366

Source: From World Bank from 1994 to 2015; from TRAI, 2016-2019; Population data from World Bank

3.5 DIGITAL DIVIDE

Diffusion of ICT: Digital Divide

The debates on the ‘information gap’ started with the rising penetration of ICTs across the world. The information gap between the developed and developing countries was initially observed in the early 1990s, which gave birth to the term ‘digital divide’. Similarly, Singh (2010), described ‘digital divide’ stating that “It is a situation in which there is a substantial gap in access to the use of ICTs devices measured, for example, by the number of fixed-line phones per hundred inhabitants or number of mobile phone users; or the internet users in the population.” Hanimann and Ruedin explain the digital divide in terms of geographical digital divide-between regions, countries, and within countries.

In the context of India, the ‘digital divide’ across rural-urban and interstate is crucial. It plays a crucial role in the socio-economic development process and in changing the pattern of people’s lives. Also, in the case of India, the urban-rural and intra-state ‘digital divide’ has been discussed in the subsequent sections indicated by mobile phone and Internet subscribers.

Rural-Urban Digital Divide

It is clear from Table 3.7 that the rural and urban ‘digital divide’ indicated by tele-density (mobile phone & fixed phone) is widening over time. Although there has been an increase in rural tele-density since 2000, that increase is much less when compared to the urban tele-density. Thus rural-urban tele-density gap has increased from 7 points in 2000 to 131 in 2012 and reduced

at 106 in 2018; thereby indicating that the information revolution in the case of India is more of an urban phenomenon. This demonstrates that the success story of the Indian telecom sector is partial, and the diffusion of services is uneven in rural and urban India. The rural (59 per cent) teledensity is around one-third to that of urban (165 per cent), albeit the former is home to nearly 70 per cent of India's total population (Census, 2011). In other words, one can say that about 70 per cent of the people are covered by only 59 per cent of phone connections.

Table 3.7: Digital Divide: Tele-Density by Rural-Urban

Year	Total	Rural	Urban
2000	3	1	8
2005	11	2	27
2010	66	31	148
2011	73	35	161
2012	79	39	170
2013	73	41	147
2014	75	44	146
2015	79	48	149
2016	83	51	154
2017	93	57	172
2018	93	59	165

Source: TRAI, 2018

Internet subscribers in India have been increasing rapidly with more people from rural areas getting hooked to the wireless or mobile Internet. However, despite the recent growth of Internet connections in rural areas, there is still a difference between urban and rural areas. Only 16 per cent of Internet users are in rural areas as compared to 85 per cent in urban areas (Table 3.8). This rural-urban Internet divide is widening from 36 points in 2015 to 69 points in 2018.

Table 3.8: Digital Divide: Internet Penetration (%) by Rural-Urban

Year	Total	Rural	Urban
2005	0.5	-	-
2010	7	-	-
2011	9	-	-
2012	11	-	-

2013	14	-	-
2015	24	13	49
2016	26	13	58
2017	33	15	71
2018	38	16	85

Source: DOT, 2018

Regional (States) Digital Divide

Table 3.9 shows that the tele-density of less developed states such as Assam, Bihar, Madhya Pradesh, and Uttar Pradesh is lower than the more developed or industrialised states like Maharashtra, Gujarat, Karnataka, Kerala, Punjab, and Delhi. Further, the rural-urban divide is also more pronounced in less developed states, which is reflecting the widening gaps across states in the country as far as income and wealth are considered (Mani, 2012).

Table 3.9: Major State/UT-wise Tele-density - as on September 2019:

State/UT	Total	Rural	Urban
Andhra Pradesh	98.10	63.86	185.83
Assam	71.64	44.46	205.05
Bihar	59.72	43.97	157.10
Chhattisgarh*	66.2	42.1	137.1
Goa*	128.0	156.9	112.0
Gujarat	105.78	73.03	149.17
Haryana	98.85	73.31	140.60
Himachal Pradesh	149.16	113.77	403.32
Jammu & Kashmir	89.19	53.33	178.58
Jharkhand*	63.1	41.9	130.0
Karnataka	110.97	68.74	174.22
Kerala	128.25	79.55	272.99
Madhya Pradesh	79.28	43.64	138.61
Maharashtra	107.64	71.02	144.50
Odisha	75.86	59.67	147.55

Punjab	127.22	80.79	182.07
Rajasthan	86.68	58.86	173.08
Sikkim*	107.0	65.4	336.9
Tamil Nadu	117.21	89.62	133.22
Uttar Pradesh	68.91	49.22	133.55
Uttarakhand	111.3	72.9	199.5
West Bengal	86.09	54.57	163.50
Delhi*	185.3	225.2	183.9
	90.52	57.59	160.63

TRAI, 2019 (data for * from 2018)

Similar to the high penetration rate of telecom services, Internet subscribers rate is more in urbanised or industrialised states like Maharashtra, Gujarat, Tamil Nadu, and Karnataka compared to less developed states such as Bihar, Uttar Pradesh, Odisha, Rajasthan, West Bengal, Madhya Pradesh and Assam (Table 3.10). The growth rate of urban internet users is higher as compared to that of rural internet users.

**Table 3.10: Major State/UT-wise Internet subscribers (in a million)
Urban and Rural division (%), 2020**

Major State/UT	Total (in a million)	Rural %	Urban %
Andhra Pradesh	53.80	40.04	59.96
Assam	13.03	50.42	49.58
Bihar	43.88	57.80	42.20
Delhi	38.53	1.92	98.08
Gujarat	42.61	28.28	71.72
Haryana	16.02	38.40	61.60
Himachal Pradesh	5.60	60.80	39.20
Jammu Kashmir	4.88	35.00	65.00
Karnataka	43.68	32.80	67.20
Kerala	25.97	39.55	60.45
Madhya Pradesh	44.44	38.39	61.61

Maharashtra	57.59	38.47	61.53
Odisha	17.62	59.70	40.30
Punjab	24.69	29.90.	70.10
Rajasthan	39.21	43.94	56.06
Tamil Nadu	48.46	27.80	72.20
Uttar Pradesh(E+W)	84.13	44.15	55.85
West Bengal	30.31	48.50	51.50
India	687.62	36.00	64.00

Source: TRAI, 2020

3.5.1 Determinants of Digital Divide in India

The above analysis made it clear that there is a huge digital divide in rural urban and inter-state in India. Rural India is significantly lagging behind urban India in the use of ICTs. Similarly, many less developed states like north-eastern states, Uttar Pradesh, Bihar, Jharkhand, Orissa, Chhattisgarh, and Assam are lagging behind the more developed states such as Maharashtra, Gujarat, Karnataka in access and diffusion of ICTs technologies. India is a multi-cultural, multi-linguistic, and multi-religious country with complex socio-economic conditions. The use of ICT technology correlates with several economic and social aspects. The following are some of the critical determinants of the digital divide in India.

- **Availability of reliable ICT infrastructure:** Many villages in rural and remote India are still without electricity and other ICT infrastructure facilities.
- **Accessibility of digital Information:** Poor mobile phone connectivity and low internet access is the barrier for information access.
- **Education /digital capacity/skills:** Level of education, mainly digital literacy or skill among a large number of people, is still low.
- **Poverty:** Income disparity is very high in India, resulting in low purchasing power and denied access to smartphones, Internet connection, and other ICTs technologies.
- **Local Content and languages:** Lack of relevant information and contents in the local language is a significant issue for Indian masses. Most of the information is available in the English language, and hence, people who do not know English are likely to be deprived of information usage.

3.5.2 Implications of Digital Divide

The above discussed issues of the 'digital divide' are needed to be tackled properly, otherwise, it can pose serious socio-economic implications on those who do not have access to it. The following are some of the implications of the digital divide:

- **Low Economic Development and Social Interest:** Economic development of the country at various levels would be misbalanced, which will cause social issues.
- **High Poverty:** Inequality of income within the society.
- **Lack of Information:** Most of the people do not have access to knowledge through the Internet as they do not understand the web content, which is not written in regional languages.
- **Low level of Education/Skills:** Negative impact on basic literacy rate and advanced or higher education among people is severely affected.
- **Unemployment:** Unemployment among people who are not digitally competent is very high.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

Fill in the blanks:

1. Developing countries have Internet access rate than developed countries.
2. India stands at second behind.....in terms of subscribers in the world.
3. Three main determinants of the digital divide are
1 , 2 , 3

3.6 INITIATIVES TO BRIDGE 'DIGITAL DIVIDE' IN INDIA

In the above context, there is a need to take several steps to bridge the digital divide in India such as making ICT accessible to all, universalising education including digital education/skills, government support, and favourable policies, promoting e-governance, e-health, e-education, and others initiatives. The Ministry of Information & Communication Technology (MCIT) has taken many steps to ensure that the benefits of ICT technologies reach the common man residing in rural and remote areas of the country.

The MCIT ensured a proper regulatory framework to reign on the digital divide with the passage of the 'Information Technology Act' in the year 2000, open data initiatives, and data sharing Policy 2012, which gave a fillip

to information technology and e-Governance activities and projects countrywide. Further enactment of the 'Right of Information Bill' in 2002 and 'Right to Information Act 2004' by the Parliament in 2005 sets out the practical regime for people to secure access to information under the control of public authorities in order to promote openness, transparency, and accountability.

In the last couple of decades, there have been several initiatives taken by the public as well as private bodies to bridge the digital divide in India. Broadly we may categorise these activities under state promoted and private promoted initiatives. It is difficult to put forward all such initiatives in this Unit, however, some important initiatives are listed below.

Table 3.11: Select Public and Private Initiatives to Bridge the Digital Divide

Initiatives	Services
E-Chaupal: This was one of the largest initiatives by Indian Tobacco Company (ITC) Limited in 2000 to provide information to farmers on market prices of crops, new farm practices in the local language, etc. It was initiated in 40,000 villages through 6,500 kiosks in 10 States of India.	The information provided to farmers regarding inputs, technology, purchase, and sale of their produce at their doorsteps and purchase of farm produce from the farmers' doorsteps helped an estimated 4 million people.
Drishtee: It was initiated as a service provided to connect rural people with the rest of India by providing a mobile kiosk-based telephone and initiated in 2000 in villages of Haryana, Punjab, Madhya Pradesh, Gujarat, and Orissa.	The telephone services offered through kiosks in the project had helped the digital inclusion of rural people residing in around 2000 villages.
Kissan Call Centre (KCC): Ministry of Agriculture, Government of India initiated KCC in 2004 in 14 different locations covering all the States and UTs to answer the farmers' queries in 22 languages through a toll-free telephone number accessible through mobile phone and landlines telephone.	To answer farmers' queries, facilitate them to manage their crops better, and also make them aware of the availability of new agriculture technology. It has helped millions of farmers across the country.

Life Line India (LLI): One World initiated it, a charitable organization with Cisco in November 2006, to provide phone-based information services to farmers and provide expert advice on agriculture and animal husbandry related queries in 700 villages. All that the farmers needed to do was to dial the Life-lines India number from a community telephone.	This service provided under the LLI was not only vital for delivering prompt services to farmers but also resulted in the improvement of their crop quality and efficiency. This resulted in an increase in their profits by up to 150 per cent. The services proved the value of digital inclusion by educating rural farmers.
National E-Governance Plan (NeGP) was initiated by the Government of India in May 2006 to make all government services available to the citizens of the country via electronic media.	The services provided under the initiative covered 100 per cent electronic filing, electronic payment mechanisms, and the use of digital signature certificates for all transactions.
Digital India: This is a flagship programme of the Indian government initiated in the year 2015. Under Digital India, the government has emphasised the creation of digital infrastructure, digital literacy, and delivery of services through digital or electronic mode.	Under the scheme now villages have been connected through broadband services. Direct electronic-payment is being made for most of the beneficiaries of government schemes. The standard services centres run various citizen-centric services in rural areas such as the issuance of birth certificate, aadhar, death certificate, and others have been helping the villagers a lot.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

Fill in the blanks:

- The right to information act promotes and accountability in the working of the government.
- Under Digital India initiatives, most of the villages in the country have access to centre to get various citizen-centric services.
- KCC provides related information to rural people.
- E-Chaupal was initiated byto provide information onto the farmers.

3.7 LET US SUM UP

In this Unit, we discussed the emergence of ICT, its definition, nature, and role. We traced the access and growth of ICT globally and with particular reference to India. We discussed the major ICT technologies from fixed phones, mobile phones, and computers to the Internet and how these technologies have increased over the years.

We also looked at digital divides in detail across countries—developed and developing worlds, rural and urban areas, and regional level or state level. We have seen how the rural, underdeveloped regions are still lagging in access to ICT technologies. Finally, we have discussed the factors leading to the digital divide and initiatives taken by the government to bridge the digital divide.

Overall, one can say that ICT has become the life line at present, and India has also benefited a lot from its diffusion. However, still, a lot needs to be done to fully exploit the benefits of ICT, where the role of government, civil society, and people is equally important.

3.8 KEYWORDS

Digital Inequalities: Uneven distribution of digital infrastructure, the existence of a varying level of competencies and skills among the different segment of users to access the available digital resources, uneven availability of regional language content in digital platforms, and even the imbalance in the opportunities to implement the usage of digital media in offline mode. These factors indicate the prevalence of digital inequalities in any given social setup.

Digital Divide: The earlier version of digital inequalities, uneven distribution of ICT infrastructure, is referred to as the digital divide. However, the digital divide focused narrowly on the physical access to digital platforms, it classified the people based on binary divisions - ‘haves’ and ‘have-nots’. With greater penetration, however, and even saturation points witnessed in developed countries, the focus shifted from physical access to usage, competencies, skills, and opportunities based divisions, more specifically it is being referred to as digital inequalities.

Internet Access: It refers to the ability of an individual or an organization to connect their computing/mobile/digital devices into the Internet networks through the data connection option of either broadband connection or wireless connections. Depending on the availability of ICT infrastructure, the speed of the Internet to these computers varies from place to place. The nature and quality of these devices too determine the actual speed of Internet connectivity. Even after access to the Internet, the user’s ability or competencies or skills determine the quality of time spent or the quantum of information retrieved or shared through these network connections.

Internet Penetration: It indicates the proportion of a country’s population connected to the Internet platform. For example, in India, 51% of its

population (around 604 million) are connected as of 2020. India has the second-largest Internet users after China, yet in comparison, the penetration is not even as 65% of them are male, and 35% are female users. Similarly, the urban and rural areas also show a wide gap.

ICT Tools: Information and Communication Technology (ICT) based infrastructure such as capturing and processing devices and software, storage hardware, transmission networks are tools of communication that take place in a digital mode using binary numbers and packet system protocols over the collection of networks. For example, computers, laptops, mobiles, scanners, hard disks, optic fibres, cell phone towers, and satellites, etc.

3.9 REFERENCES AND FURTHER READINGS

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3.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Digital inequality is the latest connotative term associated with an earlier concept called the digital divide, which explains that there is uneven distribution, access, and participation of digital infrastructure in society. There are two concepts linked together in 'digital inequality' — digital refers to the technological resources pertinent to Information and Communication Technology (ICT), and inequality denotes the unevenness in accessing and participating with the ICT resources.
2. The information age began in the mid-nineteen century after the industrial age when the work process starts involving the creation, distribution, and application of information. It has evolved in phases from a society having radio as the primary source of information to television as a popular source of information and entertainment. Further, we moved to the wireless age or digital age of mobile phones, personnel computers, and the Internet. Over the years, these changes in technologies have dramatically transformed the sources and flow of information. The sizes of technological tools have reduced drastically, and the speed to operate and process data and communication of information also increased manifolds. We can identify the five following phases of ICT evolution.
3. Information and Communications Technology plays an essential role in the development process, it contributes towards national income or GDP and improves people's living standards by facilitating access to more livelihood opportunities, and global information. It promises a fundamental change in all aspects of our lives, including knowledge dissemination, social interaction, economic and business practices, political engagement, media, education, health, leisure, and entertainment.

Check Your Progress: 2

1. Lesser
2. China, Mobile
3. Significant determinants of the digital divide/inequality are - ICT infrastructure, education/skill levels, and usage patterns.

Check Your Progress: 3

1. Openness, transparency, and accountability
2. Broadband services through Common Service Centers
3. Agriculture and farm-related activities
4. Initiated by ITC, provides market prices of crops.

UNIT 4 POLICY FRAMEWORKS AND REGULATIONS

Structure

- 4.0 Introduction
- 4.1 Learning Outcomes
- 4.2 Digital Media Framework in India
- 4.3 ICT Policies of India
 - 4.3.1 The National Telecom Policy (1994, 1999 and 2012)
 - 4.3.2 The National Cyber Security Policy (2013)
 - 4.3.3 The National Digital Communications Policy (2018)
 - 4.3.4 The Net Neutrality
- 4.4 Regulatory Body
 - 4.4.1 Role and Responsibilities of TRAI
- 4.5 IT Laws and Rules
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- 4.6 Agencies Involved in Cyber Security
 - 4.6.1 Indian Computer Emergency Response Team (CERT-In)
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- 4.7 Social Media Guidelines
- 4.8 Let Us Sum Up
- 4.9 References and Further Readings
- 4.10 Check Your Progress: Possible Answers

4.0 INTRODUCTION

Any information (picture, audio, text, graphics, video, etc.) that is created and/or transmitted using digital means such as a network is referred to as digital media. The YouTube videos we watch, the Instagram and Facebook posts, the news links we read and share, the music we download and also the information we download for our college projects are all examples of digital media. These days, even television has become digital as a set-topbox is used for transmission of signals. This device converts signals into digital form and hence, the TV we watch is also digital media.

The digital media scenario in India has evolved in spurts and phases. Today, it seems to be ubiquitous and over-arching. There are about 2 million paid digital media subscribers, while 1.5 million have entirely switched over to digital media. 50% of the growth in digital subscription has happened within the year 2017. This rapid rise is attributed to the enormous telecom subscriber base of 1.19 billion. Mobile Phones are used by 650 million

people, while 363 million use broadband. About 600 million individuals are using the internet now. Average data usage has gone up to 3.9 GB/month. Smartphones have a penetration rate of 33% of which 83% are android phones.

Such a large subscriber base calls for a robust regulatory and policy framework. The Telecom Regulatory Authority of India (TRAI) has been entrusted with regulating the sector. The policy decisions regarding all matters related to information technology are taken by the Ministry of Electronics and Information Technology (MeitY). Together, these two bodies manage the policy and regulatory framework for digital media in India. Apart from this, there are several bodies and agencies that help in regulating the digital media landscape in India.

In this Unit, we shall discuss the regulatory frameworks currently existing in India to monitor and regulate the ever-growing digital media landscape, and the national and international policy frameworks related to Information and Communication Technologies (ICT).

4.1 LEARNING OUTCOMES

After reading this Unit, you will be able to:

- understand the policy and regulatory framework for digital media;
- recognise roles, responsibilities and powers of different bodies;
- examine the regulatory and policy challenges in the domain; and
- understand overlaps and lacunae in the system.

4.2 DIGITAL MEDIA FRAMEWORK IN INDIA

India is one of the largest telecommunication markets in the world [you might have gone through various data related to the penetration levels of ICT services in Indian telecom scenario in the previous Unit]. We have the highest data consumption, as more than 600 million Internet users and 400 million regular users of social media in India. Such rapid expansion of digital media promise calls for more avenues of equitable distribution of resources and provides new opportunities for work, thereby facilitating growth and development.

The scenario has become further complicated due to the emergence of Over-the-Top (OTT) content providers such as Disney-Hotstar, Netflix and Amazon Prime. These platforms have caused significant disruptions in the digital entertainment domain. There are no rules or guidelines to regulate content over online platforms. Though the government is mulling such a move, the content providers have been resisting it, citing the Right to Freedom of Expression enshrined in the Constitution. Digitisation and enhanced infrastructure have also allowed niche content to be published/aired in vernacular languages. It has become much easier to do targeted advertising as viewing has become much individualised. There are over 300 million Over-the-Top (OTT) subscribers in India as on 2019. Online gaming

and video calls have become exceedingly popular. Both traditional and new players have entered the digital media market.

Therefore, the digital media ecosystem has many stakeholders. These are players who are involved in creating, using and facilitating access to digital communication. On the one hand, there are Telecom and Internet service providers such as Airtel, Reliance Jio, Tata etc. who provide platforms and connectivity to reach consumers while on the other hand, there are production companies that create content for distribution. Therefore, any regulatory body dealing with the above must consider issues such as spectrum allocation, licensing, broadcast permits and infrastructure in addition to establishing guidelines for the production, transmission and distribution of content. It is paramount to secure the interests of consumers as well as the service providers.

The Internet policy researchers try to understand the practice of governing the Internet as a global and national infrastructure; the impact of public and private regulation on Internet-based economies, communities and cultures; and the rights, responsibilities, norms and principles invoked by users and non-users. India has always encouraged a multi-stakeholder approach to Internet governance. This signifies that the government wants to restrict the monopoly of any one player while holding them accountable for their actions. Additionally, the safeguarding of public interest in the administration of the Internet is of more significance.

The following are issues that need to be managed to ensure a democratic distribution of spectrum in India:-

- 1) Control of critical Internet resources
- 2) Setting Internet standards
- 3) Access and inter-connection coordination
- 4) Cybersecurity governance
- 5) Policy role of information intermediation
- 6) Architecture based intellectual property rights enforcement

There are also questions of data protection and privacy, copyright, security, digital literacy, net neutrality, content regulation and data regulation.

Access to information is vital for a vibrant democracy. At the moment, access to telephone and Internet is managed with the help of Internet Service Providers. This raises questions about implications for users in case the information infrastructure is manipulated by vested interests. The onus of governing this lies with the TRAI through MEITY as explained later in the Unit. The government protects the rights of content creators through the IT Act. The question of data regulation has been addressed through the Data Protection Bill. Privacy concerns are addressed within the legal and constitutional framework. Thus, Internet governance in India is a mix of all types of actors (Kerr et al., 2019).

The Government of India has envisioned a flagship 'Digital India' programme that is centred on three core areas:

- 1) Digital Infrastructure as a Utility to Every Citizen
- 2) Governance and Services on Demand
- 3) Digital Empowerment of Citizens

To further understand the roles and responsibilities of stakeholders, it is essential to understand the regulatory framework in India. As per the Indian Constitution, our administrative system is divided into Legislature, Executive and Judiciary. The legislature makes laws and rules for the country. It includes the Parliament and State legislatures. The executive (central and state governments) ensures law enforcement while the judiciary acts as a system of balance, so that powers of both legislative and executive are kept under check. It safeguards the constitutional rights of citizens. The legislature makes rules and regulations that are converted into laws by the executive. The judiciary safeguards these laws and ensures enforcement. The laws have to be complied by all, and any non-compliance is punishable by fine or imprisonment. Thus, legislative bodies make regulations, and the judiciary ensures compliance by backing them with strict laws.

In India, the Department of Telecommunication (DOT) operating under the Ministry of Communication handles all matters about policy, licensing and regulation in the telecom sector. It manages the Digital Communications Commission (DCC), Telecom Regulatory Authority of India (TRAI), Telecom Disputes Settlement and Appellate Tribunal (TDSAT) etc. The responsibility of the administration of laws related to telecom also rests with DOT. The Indian Government established the Telecom Commission in 1989 to deal with the administrative and financial matters related to telecommunications. It was rechristened as the 'Digital Communications Commission' (DCC) in October 2018. The DCC has the following functions:-

1. Formulating the policies of the Department of Telecommunications for approval of the government;
2. Preparing the budget for the Department of Telecommunications for each financial year and getting it approved by the Government; and
3. Implementation of the Government's policy in all matters concerning telecommunication.

Activity – 1

List the primary services offered as part of the Digital India programme.

Check Your Progress: 1

Note: 1) Use the space below for your Answers.

2) Compare your answers with those given at the end of this Unit.

1. What is the current scenario concerning digital media in India?

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2. Give an overview of the regulatory structure of digital media in India.

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4.3 ICT POLICIES OF INDIA

To better understand the regulatory and policy framework in the telecom sector, let us first understand the differences between policy, regulation, laws and guidelines. Policies are different from regulations as the purpose of a policy is to guide decisions. They are a statement of intent and are usually indicative in nature and broad in scope. Theodore J. Lowi, the American political scientist, proposed four types of policy, namely distributive, redistributive, regulatory and constituent (Perguntasapo, 2012). The regulatory policies define limits to behaviour and suggest fines for noncompliance while limiting the discretion of agencies.

4.3.1 The National Telecom Policy (1994, 1999 and 2012)

In India, the National Telecom Policy has been the guiding source for the telecommunication sector. Initially, the National Telecom Policy 1994 was introduced as part of the budgetary announcements to open up the telecom sectors to private players. The full-fledged National Telecom Policy (NTP) was introduced in the year 1999 and reintroduced in the year 2012. The NTP 1999 policy seeks to address the issues such as availability of spectrum for telecom services. An independent regulator, TRAI has been created for the purpose through the NTP 1999. TRAI regulates the allocation of telecom spectrum in India. It lays down rules for eligibility criteria, application, licensing and granting of spectrum frequencies. The bidding process contains multiple stages, and TRAI ensures that all interested players abide by the established standards at each stage.

During the NTP 1999 period, there were concerns regarding the declining revenues of private players involved in the telecom sector and the need to safeguard the IT sector against emerging threats. It envisaged providing affordable and effective communications for all Indians, including those based in rural, tribal and hilly areas. Improving the telecommunications infrastructure and proper management of the spectrum were the most important objectives of this policy.

The NTP 2012 document states that the “Vision is to transform the country into an empowered and inclusive knowledge-based society, using telecommunications as a platform.” The main aim is to enhance technology adoption so that sectors such as education, health etc. can be improved. As per the policy document, “NTP -2012 is an initiative to create a conducive

policy framework to address these issues and to touch the lives of all citizens and transform India.” NTP 2012 is aimed at further empowering the regulator.

4.3.2 The National Cyber security Policy (2013)

During the early period of the 2020s, the Information Technology (IT) sector has grown exponentially in India. The government aimed to be at par with global IT markets, for which a secure, robust networked environment was the need of the hour. It is essential to have a secure cyber ecosystem to gain the trust of global IT players. A multi-pronged approach was required to set up such an environment. The National Cyber Security Policy (NCSP) was designed to fulfil this need. The primary objectives of the NSCP were - creating a secure cyber ecosystem in the country, by generating adequate trust and confidence in IT systems and transactions in cyberspace and strengthening the regulatory framework to achieve this. The policy focused on creating strategies to gain information about cyber security threats at national and sectoral levels. Proposed to establish a National Critical Information Infrastructure Protection Centre (NCIIPC) to protect vital information infrastructure.

There was a lot of impetus on Research and Development of new IT products. The policy talks about protecting and safeguarding private data of citizens against cybercrime and theft. The NCSP assured to establishing mechanisms to prevent, investigate and prosecute cybercrime. The policy encourages public-private partnerships and collaborative engagements for securing cyberspace. However, the NCSP was silent on provisions to deal with risks from new technologies such as cloud computing, block chains etc. and increased use of social media networks.

4.3.3 The National Digital Communications Policy (2018)

In 2018 National Telecom Policy was reintroduced as National Digital Communications Policy (2018). It aims to accomplish the following strategic objectives by 2022:

1. Provisioning of Broadband for All
2. Creating 4 Million additional jobs in the Digital Communications sector
3. Enhancing the contribution of the Digital Communications sector to 8% of India's GDP from 6% in 2017
4. Propelling India to the top 50 Nations in the ICT Development Index of ITU from 134 in 2017
5. Enhancing India's contribution to Global Value Chains
6. Ensuring Digital Sovereignty

The broad mission of the NDCP 2018 policy is centred around three main targets - increasing Internet presence in India (Connect India), enhancing IT infrastructure (Propel India) and making online a secure one (Secure India).

Connect India: The primary vision of Connect India is to create a robust digital communication infrastructure. As part of this planned vision, various targets are fixed. These need to be achieved by 2022: (a) the government will establish a 'National Broadband Mission – Rashtriya Broadband Abhiyan' to secure universal broadband access; (b) take measures to recognise spectrum as a key natural resource for public benefit to achieve India's socio-economic goals; (c) ensure transparency in allocation and optimise availability and utilisation; (d) strengthen Satellite Communication Technologies in India; (e) ensure inclusion of uncovered areas and digitally deprived segments of society, and (d) customer satisfaction, quality of service and effective grievance redressal.

Propel India: Under this broad mission, the ICT infrastructure will be strengthened over a period time, that would facilitate all stakeholders to carry out various IT related services to the society. The main envisaged targets under Propel India are - (a) enabling next-generation technologies and services through investments, innovation, indigenous manufacturing and IPR generation; (b) catalysing investments for the digital communications sector; (c) ensuring a holistic and harmonised approach for harnessing emerging technologies, research and development; (e) promoting startups and SMEs, local manufacturing and value addition, capacity building, strengthening of PSUs and accelerating industry 4.0.

Secure India: Primarily Secure India aims to Ensure Digital Sovereignty, Safety and Security of Digital Communications. The Government aims to establish a strong, flexible and robust Data Protection Regime. Amendments have been proposed to various licenses & terms and conditions to address the data protection and security issues by formulating a policy on encryption and data retention, facilitating lawful interception of all Digital Communications and increasing awareness amongst users about cybersecurity-related issues. It will also ensure online safety of citizens by instituting a Computer Security Incident Response Team (CSIRT), and enforcing obligations on the part of service providers to report data breaches to authorities and affected users, based on specific parameters. It calls for strengthening the security audit mechanism and developing a comprehensive plan for network preparedness, disaster response relief, restoration and reconstruction.

The 'Digital Communications Commission' (DCC), formerly known as Telecommunication Commission, is to ensure effective implementation and monitoring of the National Digital Communications Policy-2018.

4.3.4 The Net Neutrality

The Department of Telecommunications has approved net neutrality rules proposed by TRAI that ban blocking, throttling, and zero-rating Internet data.

Zero-rating is the practice of providing Internet access without financial cost under certain conditions, such as by permitting access to only certain websites or by subsidising the service with advertising or by exempting certain websites from the data allowance. For example, Facebook was planning to offer certain apps such as Netflix, Amazon etc. on its platform

but by barring users from accessing Flipkart and Disney-Hotstar. This is considered monopolistic because the service provider gains an advantage in promoting their services over competing independent companies. It limits the rights of users to choose which service they want to access. Also, they might restrict the competitor to reach users of their service, thereby limiting the open market and invoking usage restrictions. This is unfair for all service provider in the telecom sector.

As per the Net Neutrality policy, every net user must have equal access to all lawful online content at the same speed and prescribed cost. The new rules prevent “any form of discrimination or interference” with data, including “blocking, degrading, slowing down, or granting preferential speeds or treatment to any content.” The violation of net neutrality will be treated as a violation of the license conditions. This is in contravention to most developing countries where consumers were offered free access to some Internet services via Facebook’s Internet.org or Free Basics project. It was considered limiting because instead of providing limited access to the whole of the Internet, Facebook provided free access to its carrier partners while charging for access to others. Free Basics was banned in India in 2016. This was done to ensure a level playing field for all online players.

Check Your Progress: 2

Note: 1) Use the space below for your Answers.

2) Compare your answers with those given at the end of this Unit.

1. What are the policies that govern the telecom sector in India?

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2. Why is the National Digital Communications Policy important? What are its main points?

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3. Explain the concept of Net Neutrality and its significance for India.

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4.4 REGULATORY BODY

The purpose of any regulation is to ensure the homogeneous application of the law. They are aimed at guiding the activities of government bodies. The legislature does not enact the regulations, but they help in carrying out the legislative intent. The bodies involved in regulatory activities propose regulations that are subsequently discussed and commented upon by members of the public. The proposed regulations are then issued as rules and become legally binding like laws. The purpose of regulation is to ensure a level-playing field for all stakeholders.

4.4.1 Role and Responsibilities of TRAI

TRAI was set up through the Telecom Regulatory Authority of India Act (1997), and it was entrusted to achieve the objectives of the New Telecom Policy 1999. Later, the body was split into TRAI and Telecom Dispute Settlement and Appellant Tribunal (TDSAT) via an amendment in the year 2000. TRAI was entrusted with the regulatory and recommendatory functions, and TDSAT handled dispute settlement.

TRAI has the following objectives:-

- 1) Increasing tele-density and access to telecommunications in the country at affordable prices;
- 2) Making available telecommunication services which in terms of range, price and quality are comparable to the best in the world;
- 3) Providing a fair and transparent policy environment which promotes a level playing field and facilitates fair competition;
- 4) Establishing an interconnection regime that allows fair, transparent, prompt and equitable interconnection;
- 5) Re-balancing tariffs so that the objectives of affordability and operator viability are met consistently;
- 6) Protecting the interest of consumers and addressing general consumer concerns relating to availability, pricing and quality of service and other matters;
- 7) Monitoring the quality of service provided by the various operators;
- 8) Providing a mechanism for funding of net cost areas/public telephones so that telecom operators discharge universal service obligations for the spread of telecom facilities in remote and rural areas;
- 9) Preparing the ground for a smooth transition to an era of convergence of services and technologies;
- 10) Promoting the growth of coverage of a radio in India through commercial and non-commercial channels; and
- 11) Increasing consumer choice in the reception of TV channels and choosing the operator who would provide television and other related services.

4.5 IT LAWS AND RULES

IT laws and rules are rules of conduct or action prescribed or formally recognised as binding or enforced by a controlling authority. While laws are specific, a regulation provides a broad framework for the laws to operate. Thus, both laws and regulations have their importance. Regulations are often enforced using laws.

For instance, the government may allow certain private players to participate in the bid for spectrum allocation, but their bidding might be contingent on adherence to specific laws. Non-compliance to these laws may lead to disqualification from the bidding process.

4.5.1 Information Technology Act 2000

As per the Indian Computer Emergency Response Team (CERT-In) 27,482 cybercrime were committed in India between January to June in the year 2017. The IT Act, 2000 is the primary law in India that deals with cyber crimes and e-commerce. It was amended in the year 2008 to address contemporary concerns. The Act lays down punishment and penalties for crimes committed using computers/new media, e.g. hacking (Section 66), identity theft (Section 66 C), violation of privacy (Section 66 E), threatening national security via the denial of access, and attempting to penetrate computer resources, ransom ware etc. (Section 66 F), publishing obscene, defamatory or sexually explicit content online (Section 67,67A), child pornography (Section 67 B), breach of privacy (Section 72) etc. Various sections deal with electronic records, signatures and safeguarding electronic transactions and e-governance measures.

The Information Technology (reasonable security practices and procedures and sensitive personal data or information) Rules, 2011 define provisions for personal information and security and what constitutes sensitive personal data.

Apart from the enactment of Information Technology (IT) Act, 2000, and setting up of CERT-In and National Critical Information Infrastructure Protection Centre (NCIIPC) under the IT Act, 2000, the Government of India has taken other initiatives such as introducing National Cyber Security policy (2013), and framework for enhancing Cyber Security (2013). The Government has also set up Indian Computer Emergency Response Team (CERT-in) to deal with cybersecurity and emergencies.

4.5.2 Role and Responsibilities of MeitY

The Ministry of Electronics and Information Technology (MeitY) was carved out of the Department of Electronics and Information Technology (DeitY) in 2016. DeitY was earlier known as the Department of Information Technology. MeitY is responsible for the formulation, implementation and MJM-028.indd 77 11-01-2021 12:10:46 78 New Media and Society review of national policies in the field of Information Technology, and it is also responsible for the overall infrastructure and policies in the IT sector.

Meit Y deals with:-

1. Policy matters relating to information technology; Electronics; and Internet (all matters other than licensing of Internet Service Provider).
2. Promotion of the internet, IT and IT-enabled services.
3. Promotion of Digital Transactions, including Digital Payments.
4. Assistance to other departments in the promotion of E-Governance, E-Commerce, E-Medicine, E-Infrastructure, etc.
5. Promotion of Information Technology education and Information Technology-based education.
6. Matters relating to Cyber Laws, administration of the Information Technology Act. 2000 (21 of 2000) and other IT related laws.

MeitY was entrusted with developing a Framework for Data Protection Law. Shri BN. Srikrishna, Retired Judge, Supreme Court of India, has been chosen as the Chairman for this Committee. The Draft Data Protection Bill has been submitted to the government for consideration. Subsequently, the Personal Data Protection Bill was introduced in the Parliament in December 2019. As per the provisions in this law, the data of individuals cannot be collected without giving specific details about its purpose. Moreover, individuals must be informed regarding the categories and time of collection of data and with whom it would be shared. All stakeholders are considered liable in case private data was leaked due to negligence of service providers (Section 43 A). The Section states that:-

“Where a body corporate, possessing, dealing or handling any sensitive personal data or information in a computer resource which it owns, controls or operates, is negligent in implementing and maintaining reasonable security practices and procedures and thereby causes wrongful loss or wrongful gain to any person, such body corporate shall be liable to pay damages by way of compensation to the person so affected.”

Punishment is specified in case any information obtained is disclosed to third parties without consent (Section 72 A). The appropriate authority needs to be informed in case the personal data is leaked and is likely to affect the individual negatively. The intermediary will have to provide details of such a leak on its website.

The law also specifies punishment in case the intermediaries do not retain personal data/information for the prescribed duration or fail to save a copy of the data on servers in India (Section 67 C). An intermediary is defined as *“any person who on behalf of another person receives, stores or transmits that record or provides any service with respect to that record and includes telecom service providers, network service providers, internet service providers, web-hosting service providers, search engines, online payment sites, online auction sites, online marketplaces and cyber cafes.”*

The intermediary shall not:

- initiate the transmission,

- select the receiver of the transmission, and
- select or modify the information contained in the transmission

The government will have the power to intercept, monitor or decrypt information (Section 69) and collect or monitor traffic data in the interest of national security (section 69 B). New notifications have been issued as part of the Temporary Suspension of Telecom Services (Public Emergency or Public Safety) Rules (2017) which allows the government to temporarily shut down phone and internet services during a “public emergency” or for “public safety.” But these powers are also considered over arching, and there are fears that the government may use these provisions to intercept or block legitimate social media handles in the garb of protecting national security.

Check Your Progress: 3

Note: 1) Use the space below for your Answers.

2) Compare your answers with those given at the end of this Unit.

1. Explain the roles and responsibilities of MeitY.

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2. Explain IT Act (2000) in detail listing how it can be further strengthened.

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4.6 AGENCIES INVOLVED IN CYBER SECURITY

4.6.1 Indian Computer Emergency Response Team (CERT-In)

Section 70B of the IT Act 2000 called for establishing Indian Computer Emergency Response Team (CERT-in) as the national cybersecurity agency. The agency is responsible for spreading awareness pertaining to cyber security issues and also to operate a helpline to manage such incidents. It deals with:-

- Collection, analysis and dissemination of information on cyber security incidents
- Forecast and issue alerts of cyber security incidents
- Take emergency measures for handling cyber security incidents
- Coordination of cyber security incident response activities
- Issue guidelines, advisories, vulnerability notes and white papers relating to information security practices, procedures, prevention, response and reporting of cyber incidents

Such other functions relating to cyber security as may be prescribed.

4.6.2 Data Protection Authority (DPA)

The Personal Data Protection Bill was introduced in 2019. It provides for the protection of personal data of individuals and establishes a Data Protection Authority for this purpose.

As per the Information Technology (reasonable security practices and procedures & sensitive personal data or information) Rules, 2011 – personal data is defined as-

- Password;
- Financial information such as Bank account or credit card or debit card or other payment instrument details;
- Physical, physiological and mental health condition;
- Sexual orientation;
- Medical records and history;
- Biometric information;
- Any detail relating to the above clauses as provided to body corporate for providing service; and
- Any of the information received under above clauses by body corporate for processing, stored or processed under lawful contract or otherwise.

The government has mandated that any legal entity who is processing, dealing or handling sensitive personal data (data fiduciary) must implement reasonable security practices and procedures such as data encryption, age verification and parental consent while processing data of children etc. The Bill also lays out the rights of the individual. These include the right to know if their data has been processed by a fiduciary, and even restrict continuing the disclosure.

The **Data Protection Authority** will take care that all intermediaries' (including government) process data as per the law. DPA will ensure monitoring and enforcement; legal affairs, policy and standard-setting; research and awareness; and inquiries, grievance handling and adjudication. It will state codes of practice, conduct inquiries and issue warnings.

4.6.3 National Cyber Coordination Centre (NCCC)

The government has planned to set up a National Cyber Coordination Centre (NCCC) to mitigate cybersecurity threats and understand cybersecurity breaches at the macro level.

4.7 SOCIAL MEDIA GUIDELINES

Guidelines are different from laws as they are not binding and enforceable. They suggest a course of action and do's and don'ts for any given situation. The government has specific guidelines for using social media to enhance the usage of social media for public outreach, for disseminating information, policy making, recruitment, generating awareness, education etc. Government departments and agencies can engage social media in any of the following manners:

- By making use of any of the existing external platforms, or
- By creating their own communication platforms
- The choice of the platform – based on the duration of engagement and type of consultation.

The 'Framework and Guidelines for use of Social Media for Government Organisations' comprises of the following six elements:

- Objective: Why an agency needs to use social media?
- Platform: Which platform/s to use for interaction?
- Governance: What are the rules of engagement?
- Communication Strategy: How to interact?
- Pilot: How to create and sustain a community?
- Institutionalisation: How to embed social media in the organisation structure?

Actively harnessing social media is now even more important, given the massive presence of citizens on the internet. As more and more people come online, social media can become a prominent tool to connect with masses, ensure transparency and accountability in administration, receive instant feedback and solve problems efficiently. It also reduces needless paperwork and helps in connecting people in remote and disadvantaged regions with the mainstream. In the absence of social media guidelines, there will be no standardisation among different government departments, and public outreach programs may become ineffective. These guidelines are vital to achieving the goal of 'Digital India'.

4.8 LET US SUM UP

The digital media ecosystem in the country is a made-up of various statutory bodies and constitutional provisions. A robust regulatory mechanism has been developed to deal with challenges to the system. However, rapid technological advancements and fast pace of IT adoption in India have made

it difficult to respond to emerging challenges. The government is trying to evolve systems that are sustainable and at-par with global practices. There is a huge need for digital literacy programs to enable a smooth transition of citizens into netizens. The telecom market needs to remain open, free and fair to allow rapid development of digital media.

The multi-stakeholder approach supports the democratic governance of the internet envisaged by the government. However, a large number of players are involved, and this often creates regulatory challenges along with confusion about the government's stand on issues. The shutting down of the internet to prevent misuse by non-state actors has come under severe backlash because it also infringes on the right to communicate freely. The content platforms such as Facebook, Google also face difficulties as they can neither monitor all the content shared on their platforms nor risk noncompliance when asked to take down a page. The end goal of the regulatory structure is to create a fair, equitable and transparent system of internet governance that is accountable and dynamic.

4.9 REFERENCES AND FURTHER READINGS

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4.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Digital media has registered unprecedented growth in the past few years. The low cost of access to data and increasing uptake of smartphones is responsible for this. The consumption of video over Over-the-Top (Netflix, Amazon Prime, etc.) and video-on-demand platforms and news in vernacular languages have caused a growth spurt in digital media.
2. The regulatory structure has not been able to keep pace with the digital media in India. The Telecom Regulatory Authority of India (TRAI) has been entrusted with regulating the sector. The policy decisions regarding all matters related to information technology are taken by the Ministry of Electronics and Information Technology (MeitY). Together, these two bodies manage the policy and regulatory framework for digital media in India.

Check Your Progress: 2

1. In India, the National Telecom Policy came in the year 1999 and was revised in the year 2012. An independent regulator, TRAI was created to address the issues such as availability of spectrum for telecom services in the year 1999. The National Cyber Security Policy that came in 2013 is aimed at safeguarding critical information technology infrastructure in the country. There is also a National Digital Communications Policy (2018) that seeks to ensure broadband access for all.
2. This is a broad-based policy dealing with all issues related to broadband and spectrum in India. It has three main parts: - Connect India is aimed towards providing Internet connectivity to all; Propel India is directed towards catalysing investments in the digital sector and creating a robust digital communication infrastructure, and Secure India deals with the protection of data and online safety.
3. As per the Net Neutrality policy, every net user must have equal access to all lawful online content at the same speed and cost. The new rules prevent “any form of discrimination or interference” with data, including “blocking, degrading, slowing down, or granting preferential speeds or treatment to any content. In data-heavy markets like India, net neutrality is important so that all stakeholders have a level playing field. The fundamental issue of access is dealt with by ensuring that one player/ISP does not get any undue advantage over the other in any way.

Check Your Progress: 3

1. The Ministry of Electronics and Information Technology (MeitY) was carved out of the Department of Electronics and Information Technology (DeitY) in 2016. MeitY deals with: Policy matters relating to information technology; Electronics; and Internet (all matters other than licensing of Internet Service Provider); Promotion of internet, IT and IT-

enabled services; Promotion of Digital Transactions including Digital Payments; Assistance to other departments in the promotion of E-Governance, E-Commerce, E-Medicine, E-Infrastructure, etc. Promotion of Information Technology education and Information Technology-based education; Matters relating to Cyber Laws, administration of the Information Technology Act. 2000 (21 of 2000) and other IT related laws.

2. The Information Technology Act, 2000 is the primary law in India that deals with cyber crimes and e-commerce. It was amended in the year 2008. The Act lays down punishment and penalties for crimes committed using computers/new media, e.g. hacking, identity theft, violation of privacy, publishing obscene, defamatory or sexually explicit content online etc. Various sections deal with electronic records, signatures and safeguarding electronic transactions and e-governance measures. It can be strengthened by quickly amending it as per changes in technology.

